Flock: Forecasting How Busy a Venue Will Be, and Counting How Many People Are There

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November 2026. Owl's November retrain is reported; the forecast's is added when scored.

Abstract

When friends make plans to meet up at a restaurant or other venue, they need an indication of the expected level of crowds in the venue. Google Maps offers a popular-times chart indicating the typical level of crowds at a given hour for a particular location, but this does not reflect the level of crowds on a particular night. This paper addresses two questions. Can AI tell how crowded a venue will be before you go? And can a credit-card-sized computer model be designed to count the number of people inside a venue using a thermal camera without identifying individuals? For the first question, my app Flock uses recent live data to correct a typical weekly crowd level pattern at each venue. Crowd levels are then assigned to one of five possible crowd levels: Quiet, Not Busy, Steady, Busy, or Packed. This prediction was tested on 119,939 instances of data that had not previously been used for this prediction. When the prediction was made one hour in advance, it correctly predicted the crowd level for the next live data provided by the data provider with a 54.9% success rate compared to a 30.2% success rate for the popular-times chart. On a 0-100 scale, this prediction also outperformed the popular-times chart, reducing the average error by 31.8% compared to the popular-times chart. This improvement largely comes from the most recent live data for each venue. On 13 October, this model switched to a simpler model that is based on the most recent live data. This also places a limit on the predictive capability of this model because for predictions made two to 24 hours in advance, where the most recent live data is farther behind, the simpler model only outperforms the popular-times chart by about three percentage points. I also tested this on real crowd data. Using 5.3 million publicly available hourly data records from 7 countries that were mostly individual counts of people, I found that using the most recent count to correct the weekly crowd level pattern resulted in the lowest one-hour error for 14 of 15 datasets. For the second question, the live data provided by the data provider are estimates and not actual crowd counts. For this question, I designed a small neural network called Owl. A neural network is a model that learns to recognize something by being shown many labeled examples instead of following rules that were written by a human. Owl learned to detect people in images from a low-resolution thermal camera. Owl runs on a Raspberry Pi, which is the size of a credit card. Owl was 94.3% correct on 300 previously unseen images of an office environment, showing either one or two people. The hand-written rule previously used to detect people (identifying warm spots) was only 40.7% correct. When tested on images from 12 other publicly available datasets, Owl was 40.4% correct compared to the 33.0% correct from the hand-written rule. However, Owl still does not outperform the hand-written rule on images containing three or more people in these combined datasets.

Keywords: crowd prediction; popular times; nowcasting; gradient boosting; thermal imaging; people counting; on-device inference; privacy by design

1 Introduction

1.1 Background and motivation

A group deciding where to go tonight wants to know whether each option will be busy when they arrive. The common answer in consumer maps is the popular-times chart: the venue's typical level for each hour of each weekday, averaged over past weeks, sometimes with a coarse live indicator. I call this weekly pattern, lightly smoothed (Section 4.1), the venue's curve. Möhring et al. studied Google's chart as a behavioural signal [11]. The curve is a strong first guess, but it cannot see the specific evening: the rain, the game two blocks away, the holiday, the week the place became popular.

I built Flock, an app for planning nights out with friends. Its crowd card shows one integer from 0 to 100, the served number, and one of five crowd levels: *Quiet*, *Not Busy*, *Steady*, *Busy* and *Packed*. The number is relative: 100 is the busiest that venue gets, and it does not count people. Whether that number is right decides whether a plan works. This project measures how much better than popular times that number can be, and what it takes.

1.2 Research question

Can AI tell how crowded a venue will be before you go? And can a credit-card-sized computer model be designed to count the number of people inside a venue using a thermal camera without identifying individuals? Sections 2 to 8 answer the first question, made testable as: for the venues and hours a working app actually serves, can a system that watches each venue's recent live readings beat the venue's own curve, and by how much? Sections 9 to 18 answer the second: can a neural network small enough for one core of a Raspberry Pi count the people in a room from a thermal camera better than the heat rule it replaces, with only counts leaving the device?

1.3 Hypotheses

H1 to H3 test the first question; Owl's benchmark and sealed tests test the second (Section 14).

- **H1.** Correcting the venue's curve with its own recent live readings reduces error against held-out live readings, on both the 0 to 100 scale and the five-level scale.
- **H2.** A model that predicts how an hour departs from the curve, using context such as weather, events and calendars but no recent history, beats the curve on cities it never trained on.

- **H3.** The value of a live reading decays within hours, so the weight it deserves falls with its age.

1.4 Contributions

For the first question, a venue's freshest live reading is worth more than any correction to its curve: adding it lifts within-10, the share of predictions within 10 points, from 31.2% to 55.2% (Section 6.6). Where the venue was read an hour earlier, the served number lands on the exact level 77.0% of the time (52.9% where the reading had changed since, against 29.1% for popular times; Sections 6.1 and 6.3). A venue's departure from its curve correlates 0.74 from one hour to the next and 0.26 across two (Section 3.2), so these are one-hour results; from two to 24 hours ahead the corrected curve keeps a smaller, steady lead (Section 6.4). The contributions are:

- A serving path, in production from 26 September to 1 November 2026, that corrects the venue's curve with an offset (its recent median gap from the curve) and a nowcast (its latest reading, weighted by age), every weight fitted strictly before the readings that score it. On 119,939 live readings it names the exact level one hour ahead 54.9% of the time, against 30.8% for the gradient-boosted model it replaced and 30.2% for popular times, and cuts the popular-times mean absolute error (MAE) by 31.8% on the provider's relative 0 to 100 readings (Section 6.1). Persistence baselines (repeating the latest reading) locate the gain there, and a simplified rule built on that finding served from 13 October to 1 November.
- Three properties of the curves on this provider's readings: the curve is poorly calibrated, observed busyness rising about half a point per point of curve; departures stay correlated for one to two hours; and within-10 accuracy partly rewards landing on multiples of 5, as the readings do (Section 3).
- An external replication on 5,321,507 hourly records of counts and occupancy estimates in 315 dataset-site series in seven countries (1,787,058 scored hours), its analysis plan fixed before any data were downloaded: on counts the curve is close to calibrated and departures last longer, and adding the latest departure to the next hour's curve still helps (Section 6.8).
- A gradient-boosted decision-tree model of the departure from the curve (the delta model) that beats predicting the mean, R^2 above zero, in three cities it never trained on (Section 6.2), with a complete account of its inputs (Section 5).
- Checks that outlast any one model: a gate that ships a model only if it beats the rule the app served from 13 October to 1 November, tests that training and serving compute the same inputs, a dataset that records where each reading came from, and a retrain test whose split rule and thresholds were committed before its data existed, with a comparator chosen later (Sections 4 and 7.2).

1.5 Outline of the paper

Sections 2 to 8 answer the first question with the forecast: related work, preliminary findings, methods, what goes into the model, results, and discussion with the November retrain. Sections 9 to 18 answer the second with Owl, the people counter that runs on Flux, Flock's venue sensor: it counts the people a thermal camera can see and sends a few numbers, never an image; Section 14.7 reports its November retrain, owl-5. Section 19 records how both systems were built, Section 20 what limits each answer, and Section 21 the measurements that would strengthen them: the forecast's retrain, and joining the two studies so that the forecast is checked against real counts. Section 22 answers both questions. Appendix A lists the choices made after seeing data.

2 Related Work

Sections 2 to 8 answer the first question.

Point-of-interest visit forecasting. The closest published task forecasts hourly visits to named points of interest (POIs). BysGNN [7] does so for 400 POIs in each of five US cities from SafeGraph data, whose hourly visits also underlie mobility network models of epidemic spread [3]. It reads the previous 24 hours of visits at every venue in a city and forecasts the next six hours, a live feed a consumer app rarely holds. Against a Historical Average baseline (the same weekday over the previous month) that its authors liken to the popular-times chart, its MAE is 40% to 54% lower across the five cities (Houston 8.860 to 4.095 visits); against last week's value at the same hour the gain is 14% to 19% (both computed from its tables). Its more often quoted 4.34% to 6.71% margin is over the strongest competing graph model. Its mean absolute percentage error of 60% to 80% shows what precision the task admits; the measure breaks down near zero [8]. Timokhin et al. [16] estimate hourly venue popularity from venue attributes and compare popular-times data with WiFi counts.

Occupancy prediction and granularity. Bollenbach et al. [2] report R^2 up to 0.87 for whole-beach visitor counts one step ahead, while section-level occupancy forecasts one to three days ahead have R^2 near zero and often below it (−1.26 to 0.42 across the best models). Following Moore et al. [12], they read R^2 (the share of variation a model captures) above 0.3 as weak, 0.5 as moderate and 0.7 as strong. My task sits at a crowd card's granularity: one venue, one hour.

Beating the trivial baseline is not guaranteed. In a beach-occupancy study [1], a feature-based random forest and a seasonal time-series model (SARIMA) both had higher MAE than naive persistence but lower root mean squared error, which weights large misses more. In weekly foot-traffic forecasting, a simple weighted average beat an LSTM neural network [9]. These results are why every candidate in this project is scored against the served number and against persistence of the latest reading, the standard naive benchmark [8], on the same rows (Sections 4.5 and 6.1).

3 Preliminary Findings

While designing the correction, I measured the curve against live readings (Section 4.2). Each finding fixed one design decision.

3.1 The curve is a poorly calibrated forecast

I joined every verified live reading to its venue's curve value for the same weekday and hour, giving 9,903 paired rows. A reading's departure from the curve flips sign as the curve rises (Figure 1). In hours whose curve value is at most 25% of the venue's daily peak, observed busyness sits 15.60 points *above* the curve (1,162 rows). In hours above 75% of the peak it sits 13.56 points *below* (5,746 rows). The middle quarters deviate by +1.68 and −4.99.

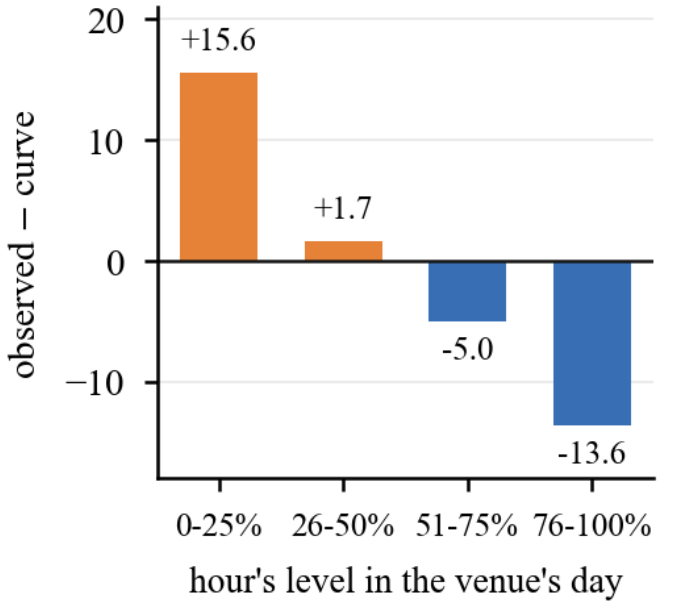


Figure 1. Observed minus curve, by the hour's level in the venue's own day (9,839 of the 9,903 paired rows).

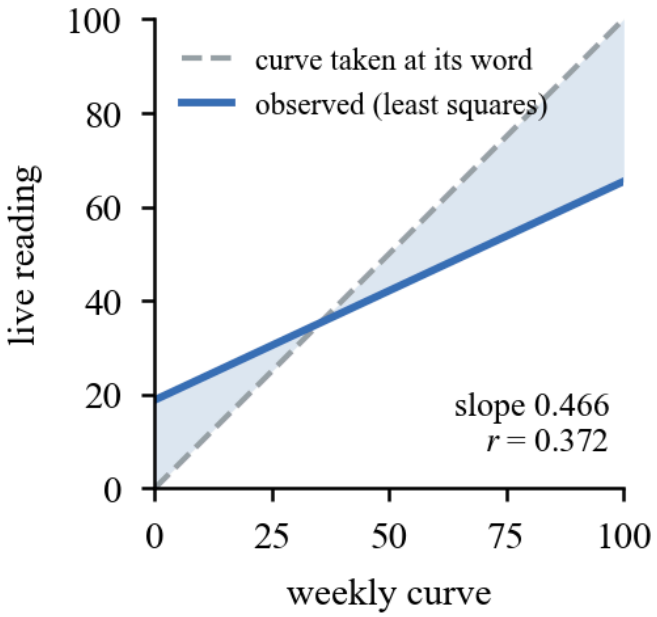


Figure 2. The least-squares fit of live readings on the curve (9,903 rows), against the identity line.

Regressing observed busyness on the curve gives *observed* = 18.82 + 0.466 × *curve*, with correlation r = 0.372 (Figure 2): each point of curve is worth about half a point of observed busyness, and the curve explains about 14% of the variance in live readings. The readings are more spread out than the curve (standard deviation 33.2 against 26.4), and the slope is low because the correlation is low. The best forecast from such a curve shrinks it toward the mean, the regression effect in prediction [5, 6]. The 30-day test window (9 September to 8 October) repeats the pattern: *observed* = 18.62 + 0.497 × *curve*, r = 0.360, with quiet hours 11.5 points above the curve and busy hours 13.9 below. The same effect explains why venues' offsets averaged about −11 (1,048 venues, 2026-09-06), which I had first read as a plain bias; on these 9,903 rows the mean departure is −6.3. **Design decision:** one global offset cannot move quiet hours up and busy hours down, so the correction works per venue and uses the latest reading; a slope that bends the curve is weighed in Section 6.6.

3.2 A venue's recent state persists, then fades

In the first week of live collection (1 to 8 September) the provider's live readings were sticky: 80.9% of consecutive-hour readings of a venue were identical, and a venue's departure from its curve correlated 0.815 across one hour, 0.487 across two and 0.203 across three (Figure 3). As the next hour's prediction, a reading one hour old landed within one level 92.0% of the time and within 10 points 85.9% (3,165 readings). Nothing in a venue's curve carries this information. **Design decision:** the latest reading should enter the served number directly, at a weight that decays on the same timescale.

The stickiness did not last. Identical consecutive readings fell to 69.3% from 9 to 25 September, 44.8% from 26 September to 2 October and 47.5% from 3 to 8 October, and over the 30-day test window the departure correlates 0.739, 0.255 and 0.158 at one, two and three hours. These data cannot tell whether the early stickiness came from the provider's refresh rate or the collector's timing.

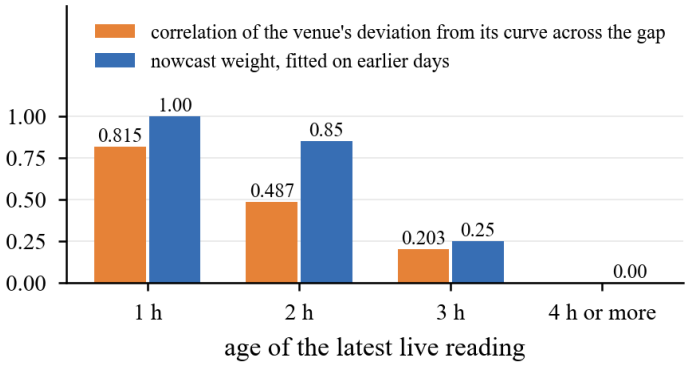


Figure 3. How a venue's departure from its curve persisted over 1 to 8 September, against the nowcast weights fitted on 1 to 5 September (Section 4.4). The fitted weights fall with age, more slowly than the correlation at two hours. Over the 30-day test the correlation at two hours fell to 0.26.

3.3 The within-10 metric rewards the grid

The provider's readings lie on a grid, or lattice, of multiples of 5, and the stored weekly value (popular times) is the provider's typical week as given, one per venue, weekday and hour, so it sits on the same grid. In a forward split, I fitted each venue's offset on its earlier readings and scored its later ones (894 venues, 3,930 later readings) at offset weights λ from 0 (no offset) to 1 (Figure 4).

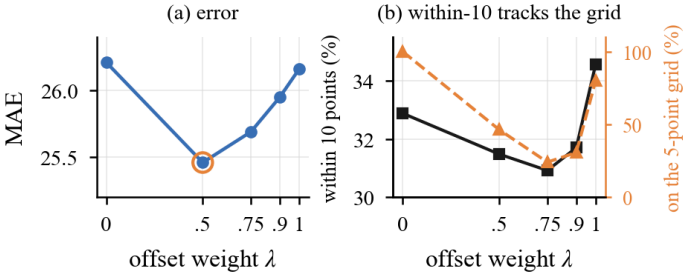


Figure 4. The forward offset test. (a) MAE is best at λ = 0.5. (b) Within-10 peaks at the two weights whose predictions land most often on the 5-point grid, not where the error is lowest.

MAE is best at $\lambda = 0.5$ (25.46, against 26.21 with no offset). Within-10 peaks at the two weights whose predictions most often land on the grid: $\lambda = 1$ (34.55%, 79.9% on-grid) and $\lambda = 0$ (32.88%, 100% on-grid). The median error is 20 at every weight. On grid-valued data, within-10 partly measures grid coincidence. **Design decision:** a within-10 comparison of a gridded and a non-gridded predictor reports the on-grid rate or projects both to the grid (Section 6.5). Production's within-10 margins in Table 5, 24 points over popular times and over curve plus offset, far exceed the variation across these weights.

4 Methods

This section defines the task and how Section 6 tests H1 to H3.

4.1 Definitions and problem statement

Busyness and crowd levels. For a venue v and an hour h, busyness $y \in [0, 100]$ is a relative score that counts no one: 100 is the busiest that venue gets. The app publishes the prediction $\hat{y}$, rounded, as the served number, and its crowd level, or *band*: *Quiet* (0 to 20), *Not Busy* (21 to 39), *Steady* (40 to 69), *Busy* (70 to 84) and *Packed* (85 to 100).

The weekly curve. Let $b(v, d, h)$ be the venue's typical value for weekday d and hour h: the venue's stored weekly value on its local clock, smoothed with its two neighbouring hours at weights 0.6, 0.2 and 0.2. The popular-times baseline is the stored weekly value itself, before smoothing, as a popular-times chart shows it. The smoothed curve is a whole number but not always a multiple of 5.

Problem. Given a venue v, a target hour h, the curve b and the live readings of v taken strictly before h, produce $\hat{y}(v, h) \in \{0, \ldots, 100\}$ and its crowd level, to minimise error against the live reading at h.

4.2 Data collection

Table 1. Data sources. The busyness provider is a commercial venue-busyness service; its scores are relative, 0 to 100 within each venue.

Source	Contribution
Weekly curves	The busyness provider's typical week for each venue: 168 hourly values on a 21-point lattice of multiples of 5. These rows define the curve.
Live readings	The busyness provider's reading for one venue at one hour on a real date, collected with a provenance label (live or provider forecast) since 2026-09-01, hourly from 3 September. About 1,380 Pennsylvania venues were scheduled every hour and Miami venues at two fixed hours a day; since late October, six cities are read hourly and 24 more in the time left. The 30-day window holds about 7,600 model-served live readings a day (realtime, with a curve above zero; Section 4.3) from 26 September to 2 October and about 4,600 a day from 3 to 8 October, after a drop on the provider's side.
Weather	Temperature, humidity, wind, condition code and rain flag at the hour, from OpenWeather.
Events	Nearest ticketed event's distance, type and attendance, and counts within 2 km, from Ticketmaster listings. For the November retrain, 633 professional and college games from TheSportsDB (2 November), refreshed daily.
Venue attributes	Category, rating, review count, price level and up to three place types, from a places listing. The venue's location is used to look up weather, daylight and neighbours and is never a feature.
Calendars	Holiday, school break, holiday eve and special nights across 29 country and city scopes.
Owner readings	A venue's own live 0 to 100 report, at training weight 0.10.
In-app reports	Users' crowd reports, verified and aggregated per venue over 28 days (the feedback family of Table 3); none existed when v2.6.0 was trained.

Procedure. Table 1 lists every source. An hourly collector reads every tracked venue's live value and stores one row per venue per hour, stamped with the venue's local clock and a provenance label. From 2026-09-25 the full sweep finished inside the hour (8 min 42 s for a 5 a.m. sweep that called 358 venues and skipped the rest as closed, about 31 minutes for a 1,321-call evening sweep); before that it stopped at its 50-minute budget. Weekly curves and context tables refresh on their own schedules. Earlier curve snapshots were not kept, so the 30-day test (Sections 4.5, 6.1, 6.3 and 6.4) scores every configuration with the curves stored at the 8 October export, while its live inputs are limited to readings that existed when each forecast was made; the replay is therefore not fully point-in-time.

Size. When the delta model was trained, the corpus held 3,912,357 rows: 3,454,955 weekly and 457,402 realtime, across 34,784 venues in 34 cities, collected between 2026-03-10 and 2026-05-18. Realtime rows were scarce: 457,402 observations over 18,442 venues is 25 per venue across 168 weekly slots (26.3 on 2026-09-06: 466,179 rows over 17,722 venues after the 2026-09-05 duplicate-venue merge).

Integrity. Unique keys enforce one weekly row per (venue, weekday, hour) and one realtime row per (venue, date, hour). Before they existed, I found and repaired three faults: in 3,536,360 rows the hour column held a list position instead of the clock hour (position 18 held the venue's midnight), there were 491,100 duplicate rows, and 63% of rows carried an impossible month-season pair. Every forecast accuracy figure in this paper postdates that repair. A later repair cleared 3,246,456 weekly rows of the weather of their collection moment; the shipped model had trained on them before it. The 457,402 realtime rows collected before provenance was recorded cannot be separated into live readings and provider forecasts, and stay marked *unknown*. "Live" in this paper means the provenance-verified readings collected since 2026-09-01 (hourly from 3 September).

4.3 Variables and controls

Table 2 lists the variables and the controls that hold every comparison fixed.

Table 2. Variables and controls.

Type	In this project
Independent variable	The serving configuration or model that produces the number (Sections 4.4, 4.5 and 5.1): weekly curve, curve + offset, curve + offset + nowcast, two persistence baselines, rule engine, delta model v2.6.0, incumbent v2.5.0 (the previous production model).
Dependent variables	Within one level, exact level, band MAE (the mean number of levels between predicted and observed), MAE, R^2, within 10 and within 15 points, bias.
Controlled: the rows	Every configuration is scored on the identical held-out readings, rebuilt through the production serving code.
Controlled: time	The offset and nowcast weights are fitted on 2026-09-01 to 09-05. The main evaluation covers 2026-09-09 to 10-08; the first test covered 09-06 to 09-08.
Controlled: geography	Miami, Tokyo and Barcelona never enter the shipped delta model's training, and each learner-comparison fold leaves out the city it scores. Temporal tests use a venue's earlier readings, never the scored reading or a later one.
Controlled: population	Scored rows must be *model-served* (realtime, curve above zero), a test made by one function shared by trainer and evaluator.

Type	In this project
Reference	A constant "Not Busy" (a score of 30, the band's midpoint), which shows how much a do-nothing answer earns on the level metrics.

4.4 System design

Three data sources feed two pipelines (Figure 5). The online pipeline served every venue with a curve from 26 September to 1 November, when production switched to model mode (Section 5.1). The offline pipeline trains the delta model; a retrain ships only by passing a two-part ship gate.

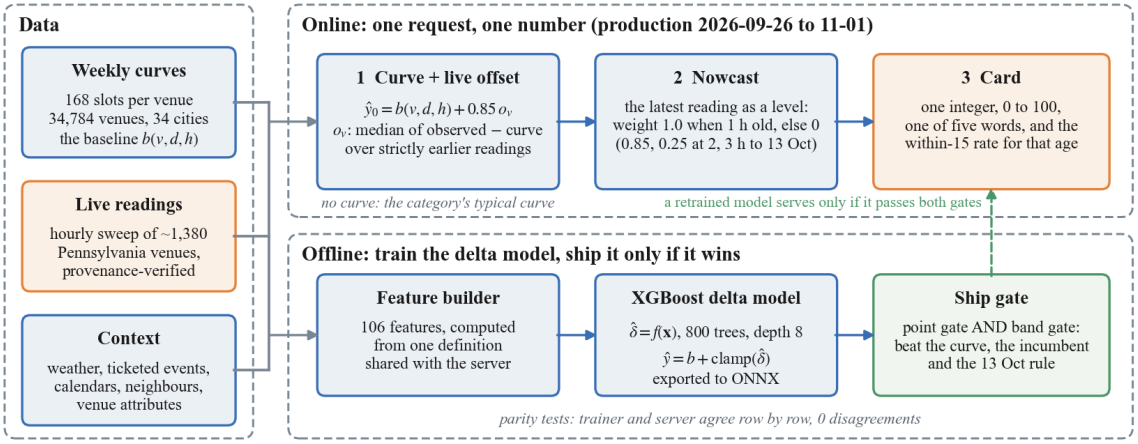


Figure 5. The system. One data column feeds an online serving path (in production 2026-09-26 to 11-01) and an offline delta model, whose retrain serves only if it passes both halves of the ship gate.

Online correction. For each venue, let the offset o_v be the median of (*observed* − *curve*) over the venue's newest 20 live readings taken within 28 days strictly before the scored hour, clamped to ±50, and 0 with fewer than two readings. Let r_v be its most recent reading from an earlier hour, and a its age in hours. The served number is

$$\hat{y}_0(v, h) = b(v, d, h) + 0.85 \cdot o_v, \quad (1)$$

$$\hat{y}(v, h) = (1 - w_a) \cdot \hat{y}_0(v, h) + w_a \cdot r_v, \quad (2)$$

each step rounded and clipped to [0, 100] as in production, with nowcast weights $w_1 = 1.0$, $w_2 = 0.85$, $w_3 = 0.25$ for readings one, two and three hours old, and $w_a = 0$ for $a \geq 4$. The offset weight and all four nowcast weights were chosen to minimise MAE on 2026-09-01 to 09-05 and held fixed through the 30-day test (served from 26 September, replayed before; Section 4.5). On 13 October the two- and three-hour weights were set to zero (Section 6.3). With $w_1 = 1.0$, a reading one hour old is served unchanged: for fresh readings the fit chose the "repeat the last reading" baseline of Section 3.2, and the offset takes over as readings age. Section 3.3's forward test, on a different split, preferred an offset weight of 0.5 to this fit's 0.85. The card publishes a confidence, an observed within-15 rate (Sections 5.2 and 6.3). Section 5 lists every serving step and input.

Offline delta model. The model predicts how an hour departs from the curve, $\hat{\delta} = f(\mathbf{x})$, from 106 input features in eleven families, listed in full in Table 3. Inputs that would reveal the answer being predicted (label leakage) are excluded, following Kapoor and Narayanan's taxonomy [10]; a weather fault found after training is recorded in Section 4.2. The learner is XGBoost [4], gradient-boosted decision trees: each tree splits the rows on one input at a time and outputs a small correction, and 800 trees up to 8 splits deep add up to the prediction (learning rate 0.01, subsample 0.9, column subsample 0.8, minimum child weight 7, L1 0.5, L2 1.0). The score is rebuilt by a fixed *score chain*, clipped to [0, 100] and rounded:

$$\hat{y} = \text{round}(\text{clip}(b + 0.5\ o_v + \text{clamp}(\hat{\delta}, \pm 50), 0, 100)). \quad (3)$$

On the geographic holdout $o_v = 0$, because the offset uses verified live readings and these spring rows predate them. The v2.6.0 ship gate scored the delta with a ±30 clamp and without rounding (Table 6); Section 6.2 also gives the model under Equation (3) exactly.

Ship gate. A candidate model must pass point and band gates.

- *Point gate:* on the held-out cities, beat the curve by 5 MAE or 0.10 R^2, not raise MAE above the incumbent's, and match the incumbent's own within-10 on the same rows.
- *Band gate:* on at least 1,000 held-out live readings over 5 or more dates in the six cities read every hour (Section 7.2), it must:
  - beat rule (b) of Section 4.5 on within one level (the date-block 95% lower bound above −1 point);
  - not be worse on within-10, by the same bound;
  - beat the weekly curve on within one level (lower bound above 0) with band MAE no worse;
  - not fall below the rule engine on within one level;
  - hold every city with 300 or more readings within 2 points of rule (b) on within one level;
  - raise MAE by at most 1 point.

4.5 Evaluation protocol and statistics

- **Time holdout.** The served number is scored on every model-served live reading from 2026-09-09, the day after its first test, to 2026-10-08, the day of the export: 119,939 readings over 30 dates at 1,754 venues (Philadelphia 63,225, Lehigh Valley 49,503, Miami 7,211). Two persistence baselines are scored beside it: rule (a), the newest earlier reading if at most 12 hours old, else popular times; and rule (b), the hour-old reading, else curve plus offset. No reading in the window informed the weights (Section 4.4). The first, three-day test, 4,183 readings from 2026-09-06 to 09-08 (Lehigh Valley 3,615, Miami 568), is reported beside it.
- **Geographic holdout.** The delta model is scored on the *gate slice*: the 67,249 realtime rows with a usable curve in the three held-out cities, a whole-city holdout [15] that is also the project's validation set (Section 6.2).
- **Cross-validation.** Leave-one-city-out over the 30 training cities, the block design recommended for spatially structured data [15]. The corpus's 34th city, Beijing, holds no v2.6.0 training rows.

- **Reproduction.** Every geographic-holdout number is recomputed from the saved model and holdout by scripts that pin seeds and thread counts, and those scripts also produce the intervals and the learner comparison (Tables 8 and 12).
- **Statistics.** Live-reading intervals come from date-block bootstraps, which recompute each result on 2,000 random redraws of whole observation dates [15]; geographic-holdout intervals use the two-way cluster bootstrap of Table 8. Tables 4 and 5 use the per-reading rows of the 30-day window, so each interval resamples 30 date blocks. Readings of one venue on one night are correlated, and the date blocks carry that: on the exact-level margin over popular times they make the interval about 4 times wider than if readings were independent, about 16 times the variance. A two-way bootstrap that resamples venues and dates together widens the intervals by a further 10% to 17%; where it changes a conclusion, Table 5 says so. In the results tables, within one level is reported with exact level and band MAE beside it, because a constant scores nearly as high on it (Section 6.1).

4.6 Reproducibility safeguards

Following common reproducibility practice [14], training pins the device, thread count and seed, and a test confirms that two runs on the same data produce bit-identical predictions. The model is exported to ONNX, a portable format for trained models, and served in-process through ONNX Runtime [13]. Each feature family is computed by the trainer and the server from one definition, and tests require them to agree on every row (the full list is in Section 19.6). An offline replay used for every live score is proven by a test to publish exactly what the serving function publishes.

5 What Goes into the Model

5.1 Two ways the number is made

From 26 September to 1 November 2026 production served the *online correction* of Equations (1) and (2): the venue's own weekly curve, its trailing offset and its latest earlier live reading. In that mode the model stays loaded but is not run, so weather, events, calendars and neighbours do not reach the number. *Model mode*, the code's default and production's since 1 November, runs the delta model v2.6.0-starling, which takes exactly 106 inputs, listed by family in Table 3. If the model files fail to load, a *rule engine* answers instead (Section 5.2, step 1).

The shipped model was trained on 1,934,988 rows from 30 cities, with weekly rows at weight 0.05 and spring realtime rows (provenance unknown) at 1.0, and with coordinates, venue identifiers, the venue's own curve and every provenance column excluded.

Table 3. Every input of the delta model v2.6.0-starling (106 features), by family, with its source at serving time, refresh rate and use in the shipped trees. *Splits*: the family's share of the 167,758 decision splits in the 800 trees; *roots*: trees that split on the family first; both counted from the shipped ONNX file. The last three rows are not inputs of v2.6.0: the venue's curve is the anchor, and the families marked † are built for the November retrain.

Family	Features	Source at serving	Refresh	Use in the shipped trees
Time (19)	hour, weekday, month; sine and cosine of each; weekend; Friday or Saturday after 18:00; lunch, dinner, late night, morning; four season flags	The venue-local clock of the hour being scored	Every request	12.4% of splits; 349 roots, 274 of them on month. Split gain: month 20.1%, its sine and cosine 4.8%
Daylight (3)	daylight hours, hours after sunset, after-sunset flag	Solar estimate from the venue's latitude and the month	Every request	11.2% of splits; 22 roots. Daylight hours alone 6.6%
Holidays (6)	holiday, school break, holiday eve, special night, special-night boost and suppress weights	Ten US federal holidays a year, US school-break windows, and 31 national holiday calendars with 1,351 dated special nights over 29 scopes (2025 to 2028)	Regenerated yearly	1.1% of splits; 55 roots, all on the suppress weight
Weather (18)	temperature, humidity, wind, rain flag; nine condition groups; temperature anomaly against a latitude-band by month climate table; four weather interactions	Current conditions keyed to about 1 km; a missing reading becomes the climate norm	Cached 30 min per location	19.5% of splits; 140 roots. Humidity, anomaly, temperature and wind each 3.8% to 4.6%
Events (16)	any event within 2 km; nearest event's attendance and its log, and distance; its type as five flags (arts, family, music, sports, other); count, total attendance and its log within 2 km; large event; event × weekend, dinner, bar	Ticketed events under way at the scored hour within 2 km; a failed lookup enters as no event	Cached 1 h	1.3% of splits; 5 roots
Category curves (2)	typical value of the venue's category at this weekday and hour; the same, refined by price and rating tier	Tables fitted on the training corpus, shipped in the model's metadata	Fixed until retrain	9.5% of splits; 17 roots
Neighbours (2)	log count of venues with a curve within about 830 m; their mean curve value at this weekday and hour, the venue itself excluded	Stored weekly curves of nearby venues	Cached 24 h	13.8% of splits; 33 roots. Second and third most split-on features (7.1% and 6.7%)
Venue (5)	category (13 classes), price level, rating, review count and its log	The venue's places listing; missing values take the corpus median, or 0 reviews	Every request	20.0% of splits; 95 roots. Log review count is the most split-on feature (9.3%)
Place types (30)	one flag for each of the 30 most common place types, from the venue's first three types	Same listing	Every request	9.4% of splits; 84 roots
Feedback (4)	mean crowd report, log report count, has-report flag, mean error of past predictions	Per-venue aggregate of verified in-app reports from the last 28 days	Cached 1 h	Never split on: no verified report existed at training time
Row type (1)	realtime or typical-week row	Always 1 at serving	Constant	1.9% of splits, 0 roots; split gain 6.1%; dropped for the retrain

Family	Features	Source at serving	Refresh	Use in the shipped trees
Own curve (0)	refused as an input	The busyness provider's typical week, smoothed 0.6 / 0.2 / 0.2	Rebuilt after each hourly sweep	The model predicts the departure from this curve, and the served number adds that departure back
Live readings† (5)	last earlier reading's departure from the curve and its age; trailing offset and its count; the curve one hour earlier	The venue's own live readings strictly before the scored hour	Hourly	47% of a candidate's split gain in the September rehearsal
Sports games† (6)	a tracked team plays today; number of games; evening game; home game; distance to the home arena; arena within 10 km	Public sports schedules, for venues within 60 km of a tracked arena	Daily	0.02% of split gain in the rehearsal

Reading the last column. Split gain, a feature's share of the reduction in training error, is recorded for the month features (month, its sine and cosine) and the row-type flag; for the others Table 3 counts splits. A split count measures how often the trees consult a feature, and it favours continuous features with many cut points over binary flags, so it ranks families roughly. Two facts in it are exact. Ten features are never split on and cannot affect the output: the four season flags, the holiday flag, the four feedback aggregates and the family-event flag. And month carries 274 of the 800 first splits, a reflection of a corpus collected in March, April and May, which the retrain answers by dropping the month family.

Training and serving, side by side. Four differences between training v2.6.0 and serving it in model mode are recorded: the row-type flag is always 1 at serving; from September on, month takes a value the spring corpus never held, and the climate table covers March to May, so the temperature anomaly is served as 0; the neighbour arithmetic became the training definition on 26 September, after the model shipped; and the shipped model's weekly rows each carried the weather of their collection moment. None touched the online correction; all four reach the number in model mode, which production has served since 1 November.

5.2 The serving path

From 13 October to 1 November, production computed the number for a venue v at hour h:

1. *Fallback.* If the model files fail to load, the rule engine answers from the hour, weekday, rating, review count, price level, place types and weather. With no positive curve value for the venue at that hour, the rule engine answered the same way until 14 October; since then the venue gets the typical curve of its category, refined by price and rating (Section 6.5).
2. *The weekly curve* $b(v, d, h)$, rounded and clipped to [0, 100].
3. *The trailing offset* o_v of Section 4.4, at weight 0.85, each reading measured against the curve at its own hour.
4. *The nowcast.* The newest live reading from an earlier hour, if at most 12 hours old, pulls the number toward itself with weight 1.0 at an age of one hour and 0 after that (before 13 October, the weights of Section 4.4).
5. *The level. Quiet* up to 20, *Not Busy* to 39, *Steady* to 69, *Busy* to 84, *Packed* from 85.
6. *The confidence.* The card publishes the within-15 rate of the arithmetic that produced the number, on the 30-day window of Section 6.1: from 13 October to 1 November, 83.0% when an hour-old reading entered and otherwise 42.7%, the within-15 rate of curve and offset alone over all 119,939 readings, not yet calibrated on readings without an hour-old reading. From 8 to 13 October it also published 40.7% and 38.6% for two- and three-hour readings, and until 8 October the three-day figures: 41.7% for curve and offset, and 87.4%, 62.8% and 43.5% for readings one, two and three hours old. The 30-day window lowered the last three, the two-hour figure most.

In model mode, production's since 1 November, step 2 becomes the curve plus the model's departure clamped to ±50. A *display map* (straight lines through 40 points pairing equally ranked v2.6.0 scores and live readings) then reshapes the scores without changing their order, and a *one-point push* moves rounded scores outside 25 to 65 one point toward the nearer end of the scale. The offset enters at weight 0.5, and the nowcast uses weights fitted on the model's own output. Live readings come from the hourly sweep of Section 4.2, at seven minutes past each hour; curves and offsets are rebuilt after each sweep.

At start-up the server refuses any model whose metadata does not record a passing ship gate, whose feature list names a feature serving cannot compute, whose ONNX input or output shape disagrees with its metadata, or whose published accuracy record is malformed, and it then serves the rule engine.

5.3 What the model never sees

No input identifies or locates a person. The only coordinates involved, the venue's, are excluded as features: latitude reaches the model only through daylight, the climate band, the special-night rule and the neighbour box. A signed-in user's account number reaches the predictor only as the key of a per-user rate limit, and the training export has no user column. In-app feedback enters as four per-venue aggregates, which the shipped model never splits on. No crowd service reads data from Flux, the sensor of Sections 9 to 18.

6 Results

6.1 H1: the served number on held-out live readings

Table 4. The served number and its comparators on 119,939 live readings over 30 dates (2026-09-09 to 10-08), none of which informed the offset and nowcast weights (fitted on 09-01 to 09-05). Rows (a) and (b) are persistence baselines; (b), found on this window, is not out of sample. Production and the baselines carry 95% date-block intervals. Bold marks the best 30-day value. Bias is mean predicted minus observed. The last row is production's first, three-day test.

Configuration	Within one level ↑	Exact level ↑	Within 10 ↑	Band MAE ↓	MAE ↓	Bias
Production to 13 Oct: curve + offset + nowcast	78.4% (77.9–78.9)	**54.9%** (53.9–55.7)	**55.2%** (53.9–56.2)	0.772	18.00 (17.73–18.28)	−1.24
(b) Hour-old reading, else curve + offset (production 13 Oct to 1 Nov)	**79.0%** (78.3–79.6)	54.7% (53.6–55.5)	54.9% (53.4–56.0)	**0.756**	**17.68** (17.38–18.03)	−1.16

Configuration	Within one level ↑	Exact level ↑	Within 10 ↑	Band MAE ↓	MAE ↓	Bias
(a) Reading up to 12 h old, else popular times	74.3% (73.4–75.0)	53.2% (52.0–54.1)	55.1% (53.6–56.2)	0.866	20.05 (19.67–20.50)	+0.48
Curve + offset, no nowcast	70.5%	34.4%	31.2%	1.058	24.55	−1.45
Popular times (stored weekly value)	64.8%	30.2%	30.9%	1.155	26.40	+4.89
Rule engine	60.6%	22.1%	19.6%	1.287	29.63	+3.95
Before: v2.6.0 as served (map on)	58.3%	30.8%	26.9%	1.337	30.94	+13.94
Constant "Not Busy" (reference)	*75.2%*	*14.8%*	*24.5%*	*1.270*	*28.37*	*−11.82*
Production on the first test (4,183 readings, 09-06 to 09-08)	*78.8%*	*56.0%*	*53.8%*	*0.735*	*17.06*	*−0.98*

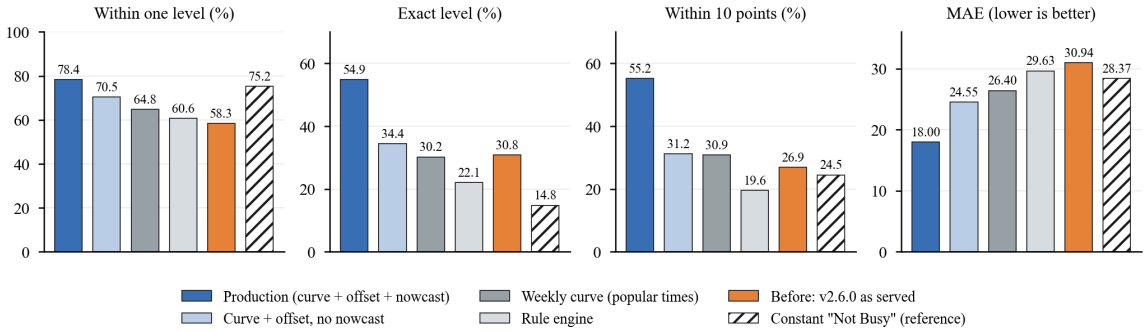


Figure 6. The served number against its comparators on 119,939 live readings over 30 dates (9 September to 8 October 2026).

Against the configuration it replaced (Table 4, Figure 6), MAE fell 42% and the bias from +13.9 to −1.2; against popular times, one hour ahead and on the provider's 0 to 100 readings, MAE is 31.8% lower and the exact level 24.7 points higher. The constant row sets the floor: real venue-hours peak at both ends of the scale (36.6% *Quiet* and 17.0% *Packed*), and "Not Busy" sits within one level of three of the five levels, so a constant scores 75.2% within one level while naming the exact level 14.8% of the time (band MAE 1.270 against 0.772). It tied production on within one level in the three-day test; over the month production clears it by 3.1 points.

Across the month. The lead over popular times holds in every period and weekday. Production names the exact level 52.5% of the time from 9 to 25 September (39,000 readings, before the online correction went live), 55.7% from 26 September to 2 October (53,287, live) and 56.7% from 3 to 8 October (27,652, after the provider's live readings thinned; Section 7.2), against 29.2% to 32.2% for popular times. By weekday it runs from 54.4% to 56.7%, Friday and Saturday included, against 28.0% to 32.5%. Part of the rise reflects more hour-old readings as the sweeps sped up (35.3%, 55.5% and 52.6% of readings); holding that share at the month's average, the three periods' rates are 59.0%, 53.0% and 54.8%. Miami is read at two fixed hours a day, eight hours apart, so its served number is curve plus offset alone; the nowcast's gain is a Pennsylvania result (112,728 readings).

Table 5. Production minus each comparator on the same 119,939 readings, in points, with 95% date-block intervals over the 30 dates. Negative MAE margins favour production. † The interval includes zero under the two-way venue × date bootstrap (exact −0.05 to 0.58; within 10 −0.05 to 0.69).

Production minus	Within one level	Exact level	Within 10	MAE
(b) Hour-old reading, else curve + offset	−0.58 (−0.79 to −0.30)	+0.24 (0.03 to 0.51)†	+0.30 (0.02 to 0.65)†	+0.32 (0.15 to 0.44)
(a) Reading up to 12 h old, else popular times	+4.1 (3.6 to 4.7)	+1.7 (1.3 to 2.2)	+0.1 (−0.1 to 0.4)	−2.05 (−2.31 to −1.85)
Popular times (stored weekly value)	+13.6 (12.3 to 14.7)	+24.7 (23.1 to 25.8)	+24.3 (22.3 to 25.7)	−8.40 (−9.02 to −7.70)
Curve + offset, no nowcast	+7.9 (6.9 to 8.7)	+20.5 (18.9 to 21.7)	+24.1 (22.0 to 25.5)	−6.54 (−6.96 to −6.05)
Rule engine	+17.8 (16.7 to 18.6)	+32.8 (31.3 to 33.9)	+35.6 (34.0 to 36.8)	−11.63 (−12.08 to −11.02)
Before: v2.6.0 as served	+20.0 (18.7 to 21.5)	+24.1 (22.6 to 25.4)	+28.3 (26.5 to 29.6)	−12.93 (−13.72 to −12.14)
Constant "Not Busy"	+3.1 (1.1 to 5.1)	+40.1 (39.0 to 41.0)	+30.7 (28.9 to 32.0)	−10.37 (−11.33 to −9.45)

Every Table 5 interval outside rows (a) and (b) excludes zero, under the two-way bootstrap as well, including the within-one-level margins the three-day test could not settle. **H1 is supported.**

Where the accuracy comes from. Where the venue was read in the hour before (readings 41 to 77 minutes apart, 10th to 90th percentile), production served that reading at full weight. Those rows, 48.3% of the window, carry 92% of the exact-level gain over popular times and 96% of the MAE gain; on 24.7% of all readings the value was unchanged from the hour before and the served number reproduces it exactly. The persistence baselines agree. Rule (a), the latest reading or else popular times, reaches 53.2% exact. Rule (b), the hour-old reading or else (on 52% of readings) curve plus offset, is level with production on the exact level and ahead within one level (0.6 points) and on MAE (0.32); the difference is the two- and three-hour nowcast weights (Section 6.3). Over all 119,939 readings, the smoothing and the offset together lift within one level by 5.7 points over popular times and cut MAE by 1.85.

The gain holds where the value moved. On the 59,681 readings whose value differed from the venue's previous reading, production reaches 41.0% exact against 30.0% for popular times; on the 28,325 of those with an hour-old reading, 52.9% against 29.1%.

6.2 H2: the delta model on cities it never trained on

Table 6. Point accuracy on the 67,249-row gate slice, scored as the v2.6.0 ship gate scored it: the predicted delta clamped at ±30 and not rounded, with o_v = 0 because the held-out rows predate verified live readings (display map off).

Predictor	MAE ↓	R^2 ↑	Within 10 ↑
Weekly curve alone	31.48	−0.075	19.2%
Incumbent v2.5.0	30.77	−0.043	19.3%
Delta model v2.6.0	**29.42**	**+0.040**	**20.7%**

The model beats the curve by 2.06 MAE (6.5%) and 0.115 R^2, and the incumbent by 1.35 MAE. R^2 crosses zero: this is the first version to beat the labels' mean on these cities, a best-of-several result on the validation set. The gain holds in every held-out city (Figure 7): 2.26 MAE in Miami, 2.21 in Barcelona and 1.72 in Tokyo (within-10 23.6%, 20.9% and 16.9%). Every number in this section reproduces from the saved model and holdout (the incumbent from its tagged commit).

These cities, the project's holdout since March, gated every version from v2.2 on, and the serving clamp and one-point push were fitted on the first 30% of their dates, so they are its validation set. The November retrain adds a time holdout (Section 7.2).

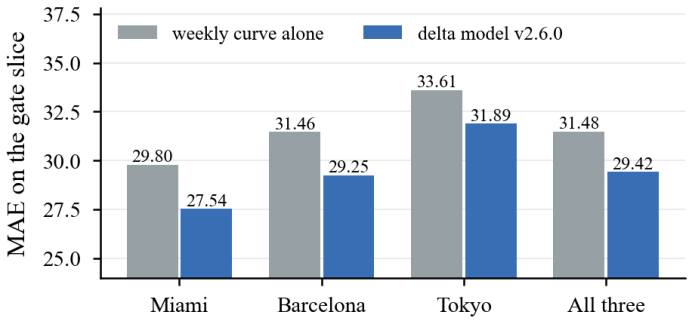


Figure 7. Delta model against the weekly curve on each held-out city of the gate slice.

On the same rows, measured on the five-level scale (Table 7), the model wins on every column: +2.6 points of exact level, +4.3 within one level, and 0.086 fewer levels of error on average.

Table 7. Level accuracy on the gate slice, with the delta clamped at ±50 and not rounded, and a score of exactly 20 counted as *Not Busy* (display map off). Production counts 20 as *Quiet*; on that ladder the gains are +2.5 exact, +4.4 within one level and 0.089 fewer levels.

Predictor	Exact ↑	Within one ↑	Band MAE ↓	MAE ↓
Weekly curve	22.9%	57.7%	1.315	31.48
Delta model	**25.5%**	**62.0%**	**1.230**	**29.35**

Scored as Equation (3) writes it, with the ±50 clamp and rounding, the model reaches MAE 29.35, R^2 +0.042 and within-10 21.7%, a point above Table 6, which clamped at ±30 and did not round.

Table 8. The delta model's margin over the weekly curve on the gate slice, with 95% intervals from a paired two-way cluster bootstrap (2,000 resamples, stratified by city) that resamples venues and city × month × weekday blocks. MAE and within 10 are scored as in Table 6, the level metrics as in Table 7.

Held-out rows	MAE	Within 10	Exact level	Within one level
All three cities	−2.06 (−2.75 to −1.44)	+1.51 (0.44 to 2.65)	+2.61 (1.55 to 3.79)	+4.31 (2.93 to 5.80)
Miami	−2.26 (−3.50 to −1.13)	+1.87 (−0.01 to 3.93)	+2.62 (0.65 to 4.78)	+4.83 (2.39 to 7.70)
Barcelona	−2.21 (−3.22 to −1.27)	+2.54 (0.38 to 4.87)	+4.27 (2.19 to 6.55)	+3.68 (1.79 to 5.76)
Tokyo	−1.72 (−2.69 to −0.83)	+0.50 (−0.66 to 1.85)	+1.71 (0.46 to 3.23)	+3.98 (2.01 to 6.17)

The holdout has no venue or date column, so I recovered both clusters. Venues come from static attributes: the daylight feature, an exact function of latitude and month, identifies 2,644 venues, none holding more than one row per weekly slot. City × month × weekday is the finest date proxy the rows carry. The dependence is temporal: date blocks widen the MAE interval 7.4 times over treating rows as independent, where venues alone widen it 1.4 times. R^2 gains 0.115 (0.079 to 0.154), band MAE falls 0.086 (0.058 to 0.116), and against the incumbent v2.5.0 MAE falls 1.35 (1.00 to 1.72). Every margin excludes zero overall and in every city except within-10 in Miami and Tokyo, so the claim for H2 rests on MAE and the level metrics.

Scored out of fold by models that never saw their city, the 369,076 served realtime rows give MAE 27.54, R^2 0.127, within-10 22.8% and within-15 33.3%. Refit the same way and scored 15 days forward within the 30 training cities (103,254 realtime rows), the model scores MAE 29.02 against the curve's 30.23 (within-10 20.7% against 20.4%), and R^2 0.078 against 0.016. The gain is smallest at dinner (17:00 to 20:00) and in the evening (21:00 to 23:00), 0.96 and 0.87 MAE.

Against prior work, the model's 6.5% gain uses no recent history, where BysGNN's larger reductions read the previous 24 hours at every venue [7]. The model's R^2 of +0.040 is positive at a granularity where the nearest peer-reviewed occupancy study's section-level forecasts a day or more ahead have R^2 near zero and often below it [2]. **H2 is supported on the held-out cities.** The model learned from spring rows of unknown provenance, and on verified live readings the online correction of Section 6.1 gives the stronger number, so production served it from 26 September to 1 November (Sections 6.1 and 19.5).

6.3 H3: the value of a reading by its age

The served number's accuracy is set largely by how recently the venue was read (Figure 8). Over the 30-day window, 48.3% of readings (57,900 of 119,939) had an hour-old reading, and on those the served number lands within 15 points 83.0% of the time, within one level 88.6% and on the exact level 77.0%, with an MAE of 9.99. With a reading two hours old, within-15 falls to 40.7% (14,778 readings), and with one three hours old to 38.6% (7,794), level with curve and offset on the same rows (39.7% and 38.7%); with no reading in the last 12 hours it is 44.4%. On the three-day test the two-hour figure was 62.8%, the card's confidence until 8 October (Section 5.2, step 6); over the month nearly all of a reading's value sat in its first hour.

The month also refined the weights. On the 14,778 readings whose latest reading was two hours old, curve plus offset alone beat the nowcast at weight 0.85 by 2.73 MAE (25.83 against 28.56, interval 1.41 to 3.63) and 4.5 points within one level (2.4 to 6.1), and the nowcast was no better than popular times on MAE (27.34); its weight had been fitted on 1 to 5 September, when consecutive readings were identical four times in five. At three hours the 0.25 weight made no measurable difference. **H3 is supported:** the decay in accuracy follows the decay

in correlation of Section 3.2, and is steeper. On 13 October production's two- and three-hour weights were set to zero, so it served rule (b), the retrain gate's comparator, until 1 November (Section 7.2).

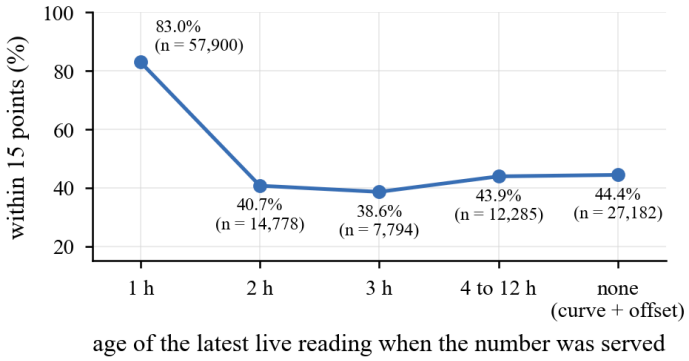


Figure 8. Accuracy by the age of the latest reading, over the 30-day window (119,939 readings). From 13 October to 1 November the card's confidence was the one-hour rate when an hour-old reading entered, and otherwise 42.7% (Section 5.2, step 6).

6.4 How far ahead

Beyond the one-hour headline, each of the 119,939 readings was forecast again as if from *h* hours earlier, using only the readings that existed then (Table 9). Two or more hours ahead the latest reading is too old to use, and rule (b) serves curve plus offset. That number stays ahead of popular times at every horizon to a day: by 3.0 to 3.4 points of exact level, 4.7 to 4.9 points within one level and 1.2 to 1.4 points of MAE, with date-block intervals that exclude zero. Within 10 is the exception (0.6 to 0.8 points lower), the grid effect of Section 3.3. The large gains are one-hour gains; the smaller, steady lead is what a group checking the card hours in advance received under rule (b).

Table 9. Accuracy by forecast horizon on the same 119,939 readings, with rule (b), production's rule from 13 October to 1 November: the hour-old reading at full weight when the forecast is made one hour ahead, and curve plus offset otherwise, using only readings from hourly slots at least *h* hours before the scored hour. Persistence is the newest of those readings, at any age; at 24 hours it is often the same hour the day before.

Hours ahead	Served exact ↑	Served within one ↑	Served MAE ↓	Popular times exact	Popular times MAE	Persistence exact
1	**54.7%**	**79.0%**	**17.68**	30.2%	26.40	53.2%
2	33.6%	69.7%	25.02	30.2%	26.40	32.3%
3	33.5%	69.6%	25.09	30.2%	26.40	30.3%
6	33.5%	69.6%	25.13	30.2%	26.40	28.4%
12	33.4%	69.5%	25.15	30.2%	26.40	28.5%
24	33.2%	69.5%	25.18	30.2%	26.40	33.1%

6.5 What coverage is worth

Coverage means a venue has its own curve. A venue with none fell to the rule engine until 14 October. Since then it gets its category's typical curve, refined by price and rating, which beat the rule engine on every scored slice: on 75,715 Philadelphia readings at 557 venues with each venue's own curve withheld, MAE 26.78 against 29.84 (−3.06, date-block interval −3.42 to −2.73) and the exact level 28.7% against 21.1% (+7.6, 6.9 to 8.6), with the same direction in the Lehigh Valley and Miami. The comparison below predates that switch.

A delta model cannot score venues without a curve, so I compared the model path with the rule engine that served them. Both got identical inputs on 5,025 verified live readings in Pennsylvania at venues with their own curve (Table 10, Figure 9).

Table 10. The two serving paths on the same 5,025 live readings (display map on, as served then).

Path	MAE ↓	Within 10 ↑	Within 15 ↑	Median error ↓	On-grid
Model path	28.22	**31.7%**	**41.5%**	21	41.5%
Rule engine	30.04	20.3%	29.2%	27	19.8%
Stored curve alone	**26.61**	30.0%	**41.5%**	**20**	100%

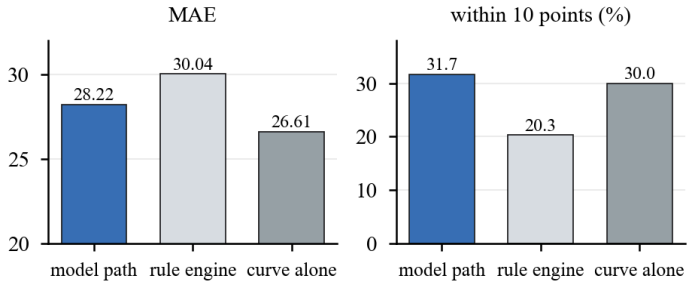


Figure 9. The model path and the rule engine on the same 5,025 live readings, with the curve alone for reference.

Through the model path, coverage is worth 1.82 MAE, 11.4 points of within-10 and six points of median error. Events were disabled in both paths, so their value is left unmeasured. Projected to the grid (Section 3.3), coverage is still worth 11.2 points of within-10, so the result is not a grid effect. The stored curve beats the model path on MAE by 1.61 while losing on within-10: on these bimodal labels the two metrics reward different predictions (Section 7.1).

6.6 Ablation experiments

Each test changes one part and holds the rest fixed.

Nowcast and offset. Table 4 separates the two corrections:

- *Smoothing and the offset* together take the popular-times row from 64.8% to 70.5% within one level, from 30.2% to 34.4% exact and from 26.40 to 24.55 MAE, and move the bias from +4.89 to −1.45. Within-10 ends where it began (30.9% to 31.2%): smoothing moves predictions off the 5-point grid and lowers it to 26.4%, and the offset lifts it back; this is the grid effect of Section 3.3.

- *The nowcast* then does most of the work: within-10 rises from 31.2% to 55.2%, and the exact level one hour ahead from 34.4% to 54.9%.

Shrinking the curve directly. The direct remedy for the poor calibration of Section 3.1 is to shrink the curve toward the mean. I measured it on the 9,903 paired rows (Table 11, Figure 10).

Table 11. Corrections of the curve's calibration on 9,903 paired rows. The per-venue rows are fitted and scored on the same readings (in-sample).

Predictor	MAE ↓	Within 10 ↑
Curve alone	26.86	31.6%
Curve + one global mean offset	25.88	28.9%
Global slope correction	25.49	21.8%
Per-venue offset, half weight (in-sample)	23.18	**35.9%**
Slope + per-venue offset (in-sample)	**22.55**	27.2%

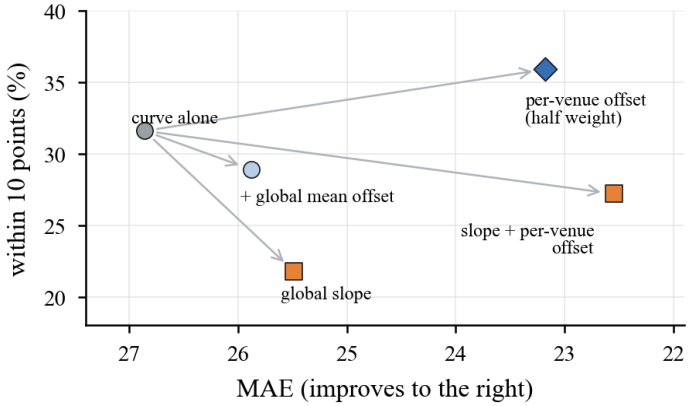


Figure 10. The corrections of Table 11. Corrections that narrow the spread (squares) improve MAE but cost within-10. The per-venue offset, the form of Equation (1), here at half weight, improves both.

Every correction that narrows the spread improves MAE but costs within-10, so the system uses the per-venue offset. A leave-one-out test over 688 venues (11,804 held-out readings) repeats the pattern: a per-venue slope improves MAE by a further 0.99 over the per-venue offset but costs 3.19 points of within-10, while the offset improves MAE by 1.56 over the curve alone (25.97 to 24.41) at a cost of 0.1 point of within-10. On 8,895 September live readings, the half-weight offset moves MAE from 25.90 to 24.53 and the exact level from 32.1% to 38.0%, on all five days and in the Lehigh Valley and Miami.

The display map. From 2026-08-28 to 09-26 the 40-point map of Section 5.2 sat between rounding and the crowd level. On 46,101 later held-out-city rows it was not fitted on, it raised within-10 by 8.4 points and MAE by 3.15. September's live readings reversed that verdict: the map cost 10.5 points of within one level (57.4% with it against 67.9% without, on 7,920 readings), and from 26 September to 1 November the served curve plus offset bypassed it (model mode applies it).

Volume against depth. A paired test scored two models on 17,322 later rows. One was trained with the scored venues' earlier history (304,000 more rows); the other had never seen those venues. The model with the extra history scored 27.63 MAE against 27.14, worse by 0.49, in a single run. A two-head variant, learning the weekly curve and the departure separately, shows the limit:

- trained on the unknown-provenance pool, it beats the shipped model by 0.88 MAE (95% interval 0.51 to 1.27);
- held to the 4,223 verified labels available at the time, it loses by 1.59 MAE (interval 0.86 to 2.45).

The limit is the number of verified labels per venue, which the live readings of Section 6.1 raise.

The learner. To test the choice of XGBoost, I held the features, target and score chain fixed and changed only the learning algorithm (the learner), rotating across the three held-out cities (Table 12). Each fold gives a learner 135,000 to 165,000 rows from two cities, far fewer than the shipped model's 1.93 million from 30, so it appears for reference. Neither boosted model was tuned; with seeds and thread counts pinned, two runs gave identical predictions.

Table 12. Four learners and the weekly curve, with the same 106 features, target and score chain (Equation (3)), each learner trained on two held-out cities and scored on the third, pooled over the 67,249 gate rows. Bold marks the best learner in each column.

Learner	MAE ↓	R^2 ↑	Within 10 ↑	Exact ↑	Within one ↑
XGBoost, the settings of Section 4.4	29.97	0.000	21.4%	25.1%	60.7%
LightGBM	**29.80**	**+0.010**	**21.7%**	**25.3%**	**61.1%**
Ridge regression	30.18	+0.003	20.3%	24.1%	59.9%
Category-by-hour mean delta	31.61	−0.095	19.7%	23.3%	57.7%
Weekly curve alone	31.48	−0.075	19.2%	22.9%	57.7%
Shipped v2.6.0 (30 other cities)	*29.35*	*+0.042*	*21.7%*	*25.4%*	*62.0%*

LightGBM beats XGBoost by 0.18 MAE (0.07 to 0.29) and ties it on within-10; ridge regression is within 0.21 MAE (−0.02 to 0.49) but loses 1.09 points of within-10 (0.54 to 1.72) and 1.01 of the exact level; a category-by-hour average is no better than the curve. The two gradient-boosted ensembles did best. The shipped model beats every learner by at least 0.45 MAE, but with 30 cities against two and far more rows, the comparison cannot separate the two effects.

6.7 The retrain, rehearsed

The November retrain (Section 7.2) will add the venue's recent readings to the model's inputs. Scored on the first test's 4,183 readings (6 to 8 September), a model trained with five live features on 3,579 earlier live readings reached 79.7% within one level, +21.2 points over v2.6.0 as served (95% interval 15.8 to 24.2). The features were the last reading's departure and its age, the trailing offset and its count, and the curve one hour earlier; they carry 47% of its split gain and are worth +10.0 points of within-10 over the candidate without them. It trailed production by 12.2 points of within-10 (interval −16.3 to −2.7) and by 1.9 MAE. Treated as independent, the 4,183 readings give intervals of 19.3 to 23.1 and −14.3 to −10.1; the date-block intervals imply variances about 5 and 10 times larger, the estimates Section 7.2's power calculation uses. The band gate refused it on the MAE guard and the three-date sample, and the within-10 criterion of Section 4.4 was added so that such a loss is refused on its own.

6.8 The findings on public real counts

Every figure above is measured against the provider's estimates. To see how counted crowds behave, I repeated the analyses on 15 datasets of real counts under an analysis plan whose pass rules were fixed and hashed (fingerprinted, so any later change would show) before any count was downloaded; two amendments, each hashed before its data were fetched, added datasets and horizons. The

datasets are hourly pedestrian counters in Brisbane, Melbourne (2019 and 2025 to 2026), Dublin, Cork, Dún Laoghaire, Leeds, Auckland, Basel and Zurich; trail counters in Cambridge, Massachusetts and Seattle; hourly entries at 61 New York subway stations; and people inside a university gym and in six reading areas of the KIT Library in Karlsruhe (Wi-Fi seat estimates the library calibrates on head-counts), used with the library's permission. Together: 5,321,507 hourly records (training spans and scoring windows) in 315 dataset-site series (Melbourne's two periods counted separately; physical sites not deduplicated) in seven countries, of which the main analysis scores 1,787,058 active hours (eight-week curve at least 10). Each site was rescaled to 0 to 100 by its training-period 99th percentile (clipped at 100), curves were built from the eight weeks before each scored day, offsets and carried departures are clamped to ±50, and intervals are week-block bootstraps (Table 13, Figure 11).

Table 13. The provider's live readings beside public real counts, one hour ahead. Real counts pool 15 datasets with equal weight: 13 count people passing a sensor and two measure people inside (a gym and a library), also shown alone. Each source's MAE is on its own 0 to 100 scale, so compare within a column. The provider's newest reading is the latest at any age; on counts every hour has one. Curve + latest departure was scored on counts only.

Measure	Provider's live readings	Real counts, 15 datasets	People inside: gym and library
Weekly curve as a forecast: slope, r	0.50, 0.36	0.92, 0.79	0.76, 0.62
Quiet hours, busy hours: observed minus curve	+11.5, −13.9	+0.3, −1.4	−0.5, −4.0
Departure correlation at 1, 2, 3 hours	0.74, 0.26, 0.16	0.74, 0.61, 0.54	0.83, 0.78, 0.73
MAE: weekly curve	26.40 (popular times)	9.58	14.36
MAE: newest reading as the forecast	21.94	10.36	8.23
MAE: curve + trailing offset	24.55	7.75	7.85
MAE: curve + latest departure	–	**6.41**	**6.29**

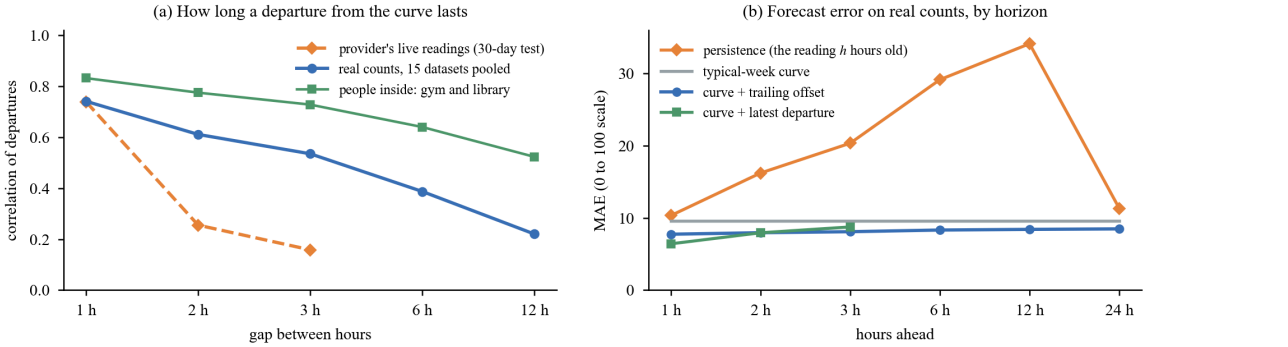


Figure 11. The provider's live readings beside 15 real-count datasets. (a) How long a venue's departure from its typical-week curve persists. (b) Forecast error on the real counts by how far ahead the forecast is made (pooled, active hours).

Part of the result carries over. A site's latest departure from its curve is its most valuable one-hour input, and the curve alone is never best: adding the latest departure to the next hour's curve gives the lowest one-hour error in 14 of the 15 datasets, 33% below the curve (MAE 6.41 against 9.58; registered without a pass rule), and the trailing offset beats the curve at every horizon to 24 hours.

The provider-specific findings do not carry over as stated. The way they fail points to the provider's readings and curves; the comparison also differs in what is counted, how curves are built and the labels' lattice. Real counts follow their curve closely (slope 0.92, r 0.79), so the raw hour-old count loses to the curve in 9 of 15 datasets; under the registered pass rules, which require 95% intervals to clear thresholds fixed in advance, the first finding (the hour-old reading beating the curve on both error and exact level) replicated in 4 datasets, and the calibration and decay findings in none. Real departures last longer (0.61 two hours on, 0.54 three hours on, against the provider's 0.26 and 0.16): the provider's value swings around its curve more widely and settles back faster, while repeating itself hour to hour.

The gym and the library, which count people inside rather than passers-by, side with the provider on the first finding: the hour-old count beats the curve by a wide margin (exact level 66.0% against 48.1%, MAE 8.23 against 14.36), the curve fits worse (slope 0.76, r 0.62), departures persist longest (0.73 at three hours), and carrying the latest departure onto the curve is better still (MAE 6.29). For a real-count source such as Flux, the evidence favours that predictor.

Every other count I could use. Nine more checks ran after the registered analysis, outside its pass rules. (1) Six KIT libraries that close at night, missing the registered rule only because Seatfinder logs nothing while closed, scored at open hours (12,248 hours): the hour-old count beats the curve (MAE 8.13 against 15.32, exact 68.3% against 47.0%), departures persist (0.93 at one hour, 0.80 at three), and the latest departure on the curve is best (MAE 5.32, exact 78.6%). (2) Five rooms at the National University of Singapore (ROBOD): on occupant counts (1,046 hours, up to 38 people) the hour-old count scores MAE 12.18 (10.66 to 13.78) against 21.63 (18.96 to 24.53) for a two-week curve, and on Wi-Fi devices (2,056 hours) 5.53 (4.92 to 6.15) against 11.77. (3) The 153 usable indoor Wi-Fi access points in the 24 buildings of the Universitat Politècnica de Catalunya's Campus Nord, Barcelona (71,926 hours, April to June 2023): the hour-old count beats the curve at 152 of 153, MAE 8.87 (8.66 to 9.08) against 16.60 (16.31 to 16.94). (4) To test whether the provider's readings follow real people, I set its typical week for 695 New York venues beside hourly entries at the nearest subway station (34 stations with at least five venues within 400 m, March to May 2026). The rhythms match closely (median r 0.74; 0.81 citywide), and at 28 of 34 stations the venues match their own station better than others (mean gain in r 0.033, interval 0.017 to 0.050), so the curve carries each block's own rhythm. Station entries are not a venue's crowd, so this checks rhythm rather than level. (5) Two Zurich series the city's open-data team suggested (CC0), through the registered engine and site rule: the footbridge counter at Cassiopeiasteg, the only one of five passing the site rule (4,526 scored hours), and the eight counting lines of the Hardbrücke

tram stop (50,588 hours). On both, the latest departure is best (MAE 7.35 against the curve's 11.77, and 5.38 against 7.05), the curve is close to calibrated (slope 0.86 and 0.97) and departures last (0.54 and 0.46 at three hours). The hour-old count beats the curve on the footbridge (9.74) and loses at the tram stop (12.36), whose commuter peaks the curve already predicts. (6) Seven more open series through the registered engine, with the KIT rule for venues that close. Where occupancy was measured, the latest departure was best on all four: 17 Aarhus libraries rebuilt from door counts (MAE 9.25 against the curve's 15.26), a Korean gym and pool from each visit's entry and exit (8.56 against 11.41), a university library's late-night head counts (5.08 against 12.91) and Wi-Fi devices in 12 Bologna libraries and museums (7.89 against 12.48). On entries it was best at six Orléans venues (14.02 against 16.58), level at a Japanese library (6.70 against 6.86) and worse at 12 Montpellier pools (17.32 against 15.65), whose departures barely persist (0.27 after an hour); there curve plus offset was best. (7) The St. Gallen and Geneva university libraries, on the same service and used with their permission, at open hours (4,308 and 11,416 hours): as at KIT, the hour-old count beats the curve (MAE 7.79 against 22.29, and 8.37 against 16.24) and the latest departure is best (5.05 and 3.91, exact 79.5% and 83.2%). (8) York's two city-centre footfall cameras (City of York Council), scored to June 2025, before the camera gaps the Council flags (10,561 hours): the latest departure is best (MAE 6.38 against the curve's 9.64), and the hour-old count does not beat the curve (10.23). (9) Toronto's island ferry (City of Toronto), as tickets scanned at the city terminal (2,060 hours): the latest departure is best (MAE 7.23 against the curve's 20.50) and the hour-old count beats the curve (7.94).

7 Discussion

7.1 Interpretation

The latest reading is the lever. Where an hour-old reading exists, using it at full weight is the strongest rule this project found (Section 6.1); where none exists, about half of venue-hours, rule (b) serves the corrected curve. On real counts, adding the latest departure to the next hour's curve is best (Section 6.8).

The delta model's gain holds in every held-out city. Blind to the venue's recent state, spring data supports a 6.5% MAE reduction and an R^2 above zero (better than predicting the labels' mean) at single-venue, single-hour granularity. On the gate slice the model's output correlates with the raw curve at 0.9638: the curve supplies the rhythm and most of the level, and the model a correction.

Readings per venue are the binding constraint. The spring corpus holds about 25 realtime rows per venue across 168 weekly slots. On 2026-09-06, 9,339 rows across 1,228 venues carried verified live provenance, a median of six each. In the one paired run of Section 6.6, 304,000 more rows of the scored venues' earlier history did not help. A reading's value lasts about an hour, so how often a venue is read matters more than how long its history is. Context does not stand in for it: in an offline test on 9 to 28 September, hourly rain, activity at Philadelphia's public bike-share docks near the venue and a local event calendar explained none of what rule (b) misses.

One number cannot sit near both ends of the scale. With 36.6% of readings *Quiet* and 17.0% *Packed*, MAE and within-10 can conflict, as Section 6.6's curve corrections show; a live reading improved both (MAE 24.55 to 18.00, within-10 31.2% to 55.2%).

Who it helps. For a group choosing where to go, the card was right more often than not one hour ahead, measured against the provider's relative readings. Flock receives only venue-level scores from the busyness provider, built from aggregated, anonymised signals; no individual's data reaches Flock.

7.2 The November retrain

The retrain, as committed in code, will learn the live features rehearsed in Section 6.7 rather than blend a reading in after the fact, give live readings at least 80% of the training weight, drop the month family and the row-type flag, and add the sports-schedule features. It is gated against rule (b) of Section 6.1, production's rule from 13 October to 1 November, which stays the comparator now that production serves v2.6.0-starling in model mode; it ships only if it wins. Three other changes in the 25 September plan (training on absolute error, dropping Beijing's rows, dropping six more features) run only after the verdict, as exploratory ablations. The export, first set for mid-October, now waits for the end of the provider's billing cycle, about 27 to 29 November, so the 14 held-out days fall in mid to late November, after the month reported here. I will update this paper with the result: a model trained on about nine weeks of verified live readings, scored on a 14-day window.

The sample. From about 23:00 on 3 October the provider answered the same calls with fewer live readings (about 4,600 a day to 8 October, 61% of the week before). The drop did not lower accuracy on the readings returned (Section 6.1); readings never returned cannot be scored. From about 16:00 on 12 October the provider returned none for about 52 hours, then a trickle from one venue from 20:52 on 14 October; the feed returned at 06:07 on 23 October. The registered rule anchors the window on the last date with live readings and needs at least 1,000 readings over 5 dates. The verdict is taken on the six cities read every hour (Philadelphia, the Lehigh Valley, Los Angeles, Seattle, Chicago and Denver); every other city, chiefly the 24 fill cities read since 30 October, is scored beside it, report only: the fill cities supply about half the readings, and the per-city guard below would otherwise bind about 30 cities.

What was fixed in advance. Hypotheses H1 to H3 were stated after the analyses that test them (Appendix A). The retrain is the first test whose hypotheses, split rule and decision thresholds were fixed in code before its data existed: the band gate's thresholds and the 14-day time holdout were committed on 25 and 26 September 2026. The comparator was chosen later, on 13 October, from the month reported here; every held-out date falls after that month. The choices above on cities, comparator, recipe and timing, and the no-curve check below, were recorded on 2 November, before the export.

- **H4.** The retrain scores higher than rule (b) on within one level over the 14-day held-out window in the six cities, and its date-block 95% lower bound is above −1.0 point.
- **H5.** It is not worse on within 10, by the same bound.
- **H6.** It beats the weekly curve on within one level with a lower bound above zero, and its band MAE is no worse than the curve's.
- **Guards.** At least 1,000 readings over at least 5 dates; neither the retrain nor production's model was fitted on the window; no city of the six with 300 or more readings falls more than 2 points behind rule (b) on within one level; within one level is not below the rule engine's; MAE rises by at most 1 point.
- **Analysis.** 2,000 date-block bootstrap resamples. The constant-answer reference, the configuration tested in this paper and production as served (model mode since 1 November) are recorded beside the verdict, each under its own label; none changes the decision.
- **Decision.** The retrain ships only if all of these hold; otherwise production keeps serving, and the result is reported either way. Before any ship, the retrain's category curves for venues with no curve of their own (Section 6.5) are scored out of sample on the window; they serve only if they beat the rule engine.

Table 14. The smallest difference the held-out window can detect at 80% power (two-sided, 5%), in points, by the number of held-out readings. Each cell gives the value with readings treated as independent, then with the variance inflated 5 and 10 times, the clustering the first-test rehearsal of Section 6.7 measured, used here as an illustration; the 30-day window measured about 16 times on the exact-level margin over popular times (Section 4.5). The assumed rates are the one-hour rates of the configuration served until 13 October, over the 30-day window. The last column gives the readings needed to detect a 1-point difference with independent readings.

Metric (assumed rate)	15,000 held out	29,000 held out	60,000 held out	Readings for 1 point
Within one level (78.4%)	1.33 / 2.98 / 4.21	0.96 / 2.14 / 3.03	0.67 / 1.49 / 2.10	26,600
Exact level (54.9%)	1.61 / 3.60 / 5.09	1.16 / 2.59 / 3.66	0.80 / 1.80 / 2.55	38,900
Within 10 (55.2%)	1.61 / 3.60 / 5.09	1.16 / 2.59 / 3.66	0.80 / 1.80 / 2.54	38,800

With about 60,000 held-out readings, a 95% interval on a within-10 difference is about ±0.6 points for independent readings and ±1.3 to ±1.8 points with 5 to 10 times inflation. With 5 to 10 times inflation, a retrain that exactly ties rule (b) would fail that bound and would need a within-10 win of about 0.3 to 0.8 points. The registered gate scores paired date-block differences, whose interval may be narrower, so the actual verdict can differ. The bound is strict on purpose: the rehearsal in Section 6.7 is the case it exists to stop.

8 Conclusions on the forecast

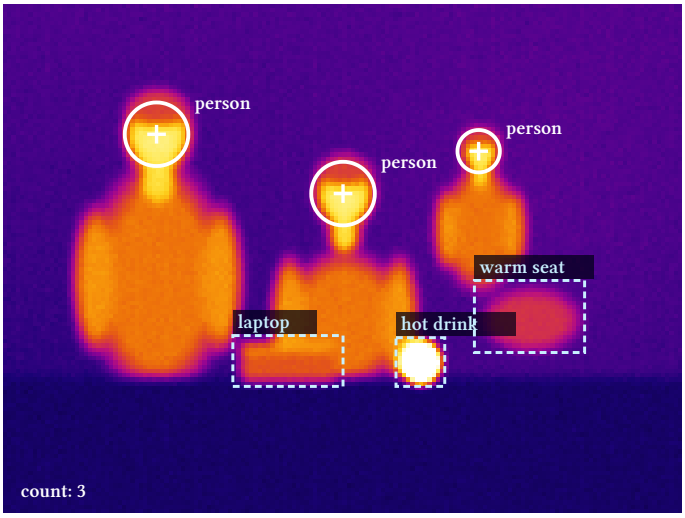
The first question asked whether AI can tell how crowded a venue will be before you go. One hour ahead, yes, and what does it is small: from 26 September to 13 October Flock served each venue's weekly curve corrected by its recent offset and a nowcast of its latest reading. Replayed on 119,939 live readings from 9 September to 8 October that its weights never saw, it named the exact crowd level one hour ahead 54.9% of the time, within one level 78.4% and within 10 points 55.2%, against 30.8%, 58.3% and 26.9% for the gradient-boosted model it replaced, with an MAE 31.8% below popular times, all on the provider's relative 0 to 100 estimates, not head counts. The three-day result it shipped on held on 29 times as many readings, on every weekday and in every period. The gain comes from fresh readings rather than a larger model: with an hour-old reading 77.0% of served crowd levels are exact, and rule (b) serves that reading at full weight. Two to 24 hours ahead the corrected curve names the exact level about 3 points more often than popular times. On 15 public datasets of counts and occupancy estimates, crowds follow their curves far more closely than the provider's readings do, and adding the latest departure to the next hour's curve gave the lowest one-hour error in 14 of them. The delta model, which production has served again since 1 November, beats the curve on three held-out cities without recent-history input (Section 6.2), but without the nowcast it loses to the weekly pattern on live readings (Section 20).

Three lessons for other systems built on typical-week curves:

- score every forecast against the latest observation as well as the curve [8]: the curve was poorly calibrated on the provider's readings (slope 0.50) and close to calibrated on real counts (0.92);
- where labels come in steps (here multiples of 5), read within-k rates beside the share of predictions on a step (Section 3.3);
- prefer fresher readings of each venue to a longer history of it: on the provider's readings a reading's value was concentrated in its first hour, though on real counts departures lasted longer.

9 Counting people on a thermal camera

Flock's forecast predicts a relative score from 0 to 100 that counts no one. Owl answers the second question: can a credit-card-sized computer model be designed to count the number of people inside a venue using a thermal camera without identifying individuals? It is a neural network that runs on Flux, a small venue sensor built around a Raspberry Pi 5 and a FLIR Lepton 3.5 thermal camera, which records heat instead of light. No Flux is installed at a venue yet, and Flock's crowd number does not read sensor data; the last sections set out how Flux's counts could check the forecast. At 160 × 120 thermal pixels Owl counts people and never identifies them; Flux reduces each frame to a count on the device and throws the frame away (Figure 12).



Camera	FLIR Lepton 3.5, 160 × 120, 57°
Input	degrees Celsius, every pixel
Model	718,550 parameters
On the sensor	main model 14–16 ms a frame, 1 Pi 5 core
Names	person + 9 kinds of warm thing
Counts	people only
Leaves the device	counts and a sound level
Footage I recorded	none

Figure 12. What Owl reads and what it reports. An illustrative 160 × 120 frame, drawn for this paper the way Owl's training generator draws a scene, shown at the camera's true resolution in the iron palette the Lepton uses. Owl puts a point on each person's head and a named box on everything else warm. Only the points are counted: this frame counts as three.

The first counter on Flux, a heat-threshold rule, counted regions warmer than the room and large enough to be a person. It failed predictably: a hand at the lens counted as a person, spread fingers as two, a coffee mug or a laptop as somebody, and a packed room merged into one warm mass. Telling these apart is a question of shape, which a learned model does well.

Sections 9 to 18 describe Owl, the model that replaced the rule, and what building it showed. Four constraints shaped the work from the first day; the first two are the question's own terms.

1. **It runs on the sensor.** One core of a Raspberry Pi 5, with no GPU and no round trip to a server.
2. **It names kinds of things and nothing more.** No identity, no face features, no tracking from one frame to the next. Only numbers leave the device: people counts and a sound level.
3. **I do not record people to train it.** Owl learns from frames I generate and from existing datasets.
4. **Every real frame is licensed.** Each source allows commercial use under its licence, or its authors gave written permission.

The contributions are a generator of thermal scenes matched to the Lepton 3.5 (Section 12.2); a 718,550-parameter network that marks each person with a point and names nine other kinds of warm object, with physics checks on the sensor and a second network, the crowd model, for busy frames (Sections 12.4 to 12.6); an evaluation against the rule on identical frames, on six real benchmark sets and sealed tests kept out of every design choice (Section 13); 15 findings from 19 full training runs plus five short trials (Section 15); and owl-5, a retrain with data from the sets those tests used, chosen and judged by rules fixed before it trained (Section 14.7).

10 Background on thermal people counting

Thermal people counting is usually studied with low-resolution ceiling thermopile arrays, like the 32 × 24 MLX90640 in PUT Thermo Presence [25], whose counting models were benchmarked on resource-constrained hardware [34] and against deep detectors on low-resolution infrared sets [39], and with higher-resolution cameras for segmentation and pose [24, 26]. Owl draws on three lines of work.

Objects as points. CenterNet [17] predicts a heat map, a grid of confidence scores (not temperatures) with a peak at each object's centre, and finds objects at the peaks. Counting then reduces to counting peaks, which is exactly the quantity a venue needs. Its loss, the penalty training minimises, is CornerNet's [32] penalty-reduced variant of focal loss [18], which handles the imbalance between a handful of peak pixels and thousands of background pixels.

Counting by density. A predicted density map [19, 20, 33] sums to the count and degrades gracefully when objects overlap, where peak finding merges two touching heads into one.

Learning from generated data. A model trained on generated frames transfers to real ones only as far as the generator covers reality. Domain randomization [21] varies everything the model should ignore so that the real world looks like one more variation. Section 15 shows where that held for Owl and where it did not.

11 The sensor and the baseline

Flux reads the Lepton 3.5 through a PureThermal 3 board over USB. The camera is radiometric: each of its 160 × 120 pixels reports a temperature in degrees Celsius, across a 57° horizontal field of view [29]. Through that lens a head three metres away is about 8 pixels wide and a hand 30 cm from the lens about 45 pixels wide.

The rule divides the frame into cells of 4 × 4 pixels and estimates the room as the frame's 20th percentile. It learns fixed warm spots such as radiators per cell over time, keeps cells warm enough to be skin, joins touching cells (diagonals included) into groups and counts groups of at least 6 cells, about one head at three metres. Its settings were tuned on a bench with a real person and against constructed scenes, and it works for well-separated people at moderate range. It has no notion of shape, which is the gap Owl fills. Throughout this paper the rule is scored on exactly the same frames as Owl.

12 Owl: method

Figure 13 shows the pipeline from training data to the sensor.

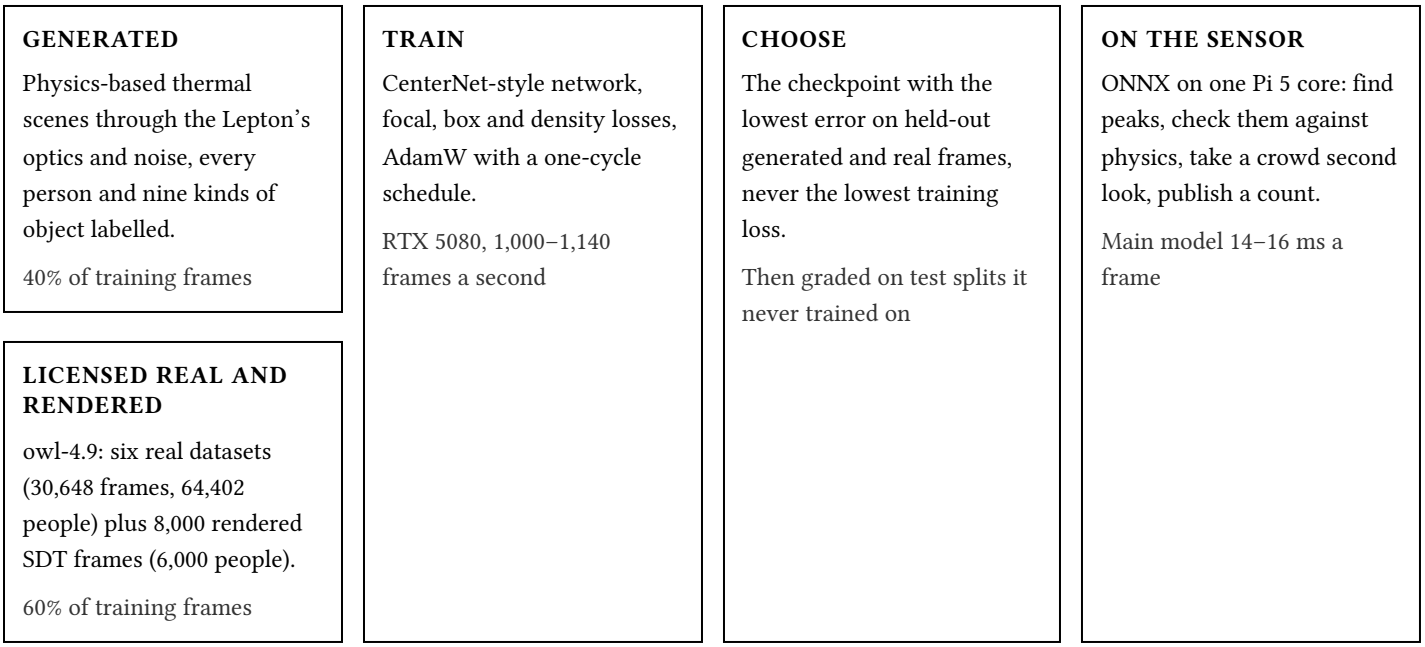


Figure 13. The pipeline. Data on the left, the deployed counter on the right. Checkpoints are selected on validation splits, and versions and settings on the real benchmark sets (Section 13). Accuracy is measured on those sets, sixteen rounds of sealed tests and generated stress scenes, and speed on the sensor itself.

12.1 Formulation

Each visible person gets one point (its head point) at the centre of the head, or of the visible body when the head is out of view. Every other warm object gets a box and one of nine kinds: hand, pet, hot drink, food, laptop, screen, heater, lamp and warm seat. Counting people is finding peaks in the person heat map. The other nine maps let the sensor's display show why something was not counted, and give the network somewhere to put what it must not count.

12.2 A generator matched to the camera

Owl's first version learned only from generated frames, and the generator stayed in every training batch after that (40% of frames for the shipped model). It draws a room at some temperature, people at

every distance and pose, and the objects that fool a threshold, at twice the camera's resolution, then averages down, blurs for the lens, and adds the camera's noise. Table 15 lists the physical model, which replaced hand-picked temperatures after the first round of real-footage tests (Finding 4).

Table 15. The physical model behind the generated frames. Values were set from published thermography of skin and clothing.

Surface	Model
Skin	33.4 °C forehead in a 22 °C room, rising 0.14 °C for each degree the room is warmer
Clothing	A garment-specific share of the way from the room to the skin; a winter coat reads barely above the room
Hair	3–10 °C above the room
Pets	Coat near room temperature, eyes and ears 35–38 °C
A seat just left	4–8 °C above the room
Rooms	Typical rooms, plus 20% of frames drawn hot at 29–34 °C
Camera	Lens blur, noise, fixed-pattern column stripes, dead pixels, vignetting, several degrees of absolute error, gain error, a crooked mount, the frame after a flat-field correction

Scenes include standing, sitting and lying people, children, hats, hoods, glasses, masks, bald heads, bare torsos, winter coats, people hidden behind tables and bars, one person partly behind another, and crowds, plus what must not be counted: hands at the lens, people reflected in glass and shiny floors, warm footprints, sun on the floor, candles, cold drinks and bulbs hotter than the camera's range.

12.3 Licensed real data

Six real thermal datasets and one set of rendered frames trained owl-4.9 (Table 16); owl-5 adds the training parts of 22 more (Section 14.7, Table 21). Two are from the Lepton 3.5 itself. Sources that store temperatures (PUT, SDT) are used in °C. Sources stored as 8-bit images carry no temperatures, so each frame is mapped onto a plausible room-to-skin range redrawn at each use (low end 16–28 °C, span 8–16 °C, brightness exponent 0.8–1.25), which stops the network treating any one mapping as truth. IISER's colour frames in the camera's iron palette are converted back to one channel by luminance, which rises monotonically along that palette from black through purple, red, orange and yellow to white.

Table 16. Real training data of owl-4.9, all kept by owl-5 (Table 21 lists what owl-5 adds). People are counted in the training split; for IISER they are labelled faces. Licences and permissions are listed in the acknowledgements.

Source	Camera and view	Stored as	Train frames	People	Use
PUT Thermo Presence [25]	MLX90640 32 × 24, ceiling	°C	8,472	24,106	MIT licence
OpenThermalPose2 [26]	Side view, indoor and outdoor	8-bit	8,846	15,849	MIT licence
AAU VAP Trimodal [24]	Wall-mounted, meeting rooms	8-bit	3,964	5,466	CC BY 4.0
SenSys 2021 set [27]	32 × 24 rooms with hot laptops	8-bit	1,239	1,577	CC BY 4.0
SDT, TU Wien [22]	**Lepton 3.5**, elevated indoor, one person or none	°C	4,800	3,600	Authors' permission
SDT rendered frames [22]	Rendered to the Lepton's look	°C	8,000	6,000	Authors' permission
Thermal Vision, IISER Bhopal [23]	**Lepton 3.5**, classroom of 12 to 15 and lab of up to 6 (up to 10 and 5 labelled faces)	Iron palette	3,327	13,804	Authors' licence and permission

Real datasets label people and nothing else, so for real frames the loss on the other nine maps is switched off. AAU, SenSys and SDT mark bodies rather than heads; for them the head point is estimated from the box and its target is spread over the pixels the head could occupy (σ = 5 px), a fix that Finding 7 explains. Test splits are the datasets' own, or, where a dataset has none, contiguous ranges in time split 60/15/25 for training, validation and test: SDT by frame range, and IISER by time within each room with ten frames dropped at every boundary, because neighbouring frames are near copies.

12.4 Network

The network sees two channels. The first is absolute, (t − 30) / 8, so a 60 °C mug can never look like skin. The second is relative to the frame's median, (t − median) / 4, so the several degrees an uncalibrated Lepton reads high or low do not move a person out of range. An encoder halves the resolution four times, to strides 2, 4, 8 and 16, building 24, 48, 72 and 96 channels (maps of learned patterns), using residual blocks [30] with dilation at the deepest stage [31, 33] to widen the view: a hand filling half the frame must be seen as one hand and not five warm blobs. A top-down path fuses the deep features back to stride 4, a 40 × 30 grid. Three output branches, or heads, read that grid: ten heat maps (bias initialised to −4.6, so training starts from "nothing here"), a box head that predicts, on a log scale, the distance from each object's centre to each edge of its box, and for the crowd model a density head, constrained to positive values and initialised near empty. The main model has 718,550 parameters and exports to ONNX as a 2.9 MB file.

12.5 Training

The loss is CenterNet's penalty-reduced focal loss on the heat maps [17, 18], with $\alpha = 2$ and $\beta = 4$ as in CornerNet [32], masked by which maps a frame actually labels; an L1 loss on the log of those edge distances, at object peaks; and, for the crowd model, squared error per density cell plus a tenth of the error in the total. I train with AdamW (weight decay 10^{-4}), a one-cycle learning-rate schedule with 5% warm-up, gradient clipping at 5, and batches of 64. Fresh runs use a peak rate of 2×10^{-3}; later versions start from the previous version, keeping every layer whose shape matches, at 2×10^{-4}. Eight CPU workers generate frames on the fly for an RTX 5080. Up to owl-4.9 the project ran 19 full training runs plus five short trials, 581,700 steps in all, which is 37.2 million training frames; owl-5 added four full runs of 20,000 steps, 5.1 million frames, with four data workers instead of eight, which changed only the speed.

12.6 Decoding on the sensor

The sensor runs the ONNX file with ONNX Runtime [13] in full precision on one core. A person is a cell scoring above 0.4 (0.35 for owl-5), on a confidence scale from 0 to 1, and at least as high as all eight neighbours. Rarer kinds, learned from fewer examples, must clear a higher bar to be named, from 1.25 times the person threshold

for hands and hot drinks to 1.75 times for pets. A person peak is dropped when a screen, laptop, heater, lamp or warm seat is more confident in the same 3 × 3 neighbourhood. Two physical checks run on a 9 × 9 patch around each person: its warmest pixel must be at least 1 °C above the room, taken as the frame's 10th percentile (Finding 3), and its median must not exceed 45 °C, which no person reaches. When five or more people are found (four for owl-5), the crowd model reads the same frame, and the published count is the highest of the main model's peaks, the crowd model's peaks and the rounded density total.

13 Owl: evaluation protocol

Checkpoints, benchmarks and sealed tests. Each run keeps the checkpoint (a saved copy of the network) with the lowest combined error on held-out generated and real validation frames, never the lowest training loss. Every version was then graded on the six test splits, the development benchmarks for choosing versions, data shares, decoding settings and the crowd hand-over point. No test frame was trained on: the training cache holds training and validation frames only, and no exact copy of a test frame. The sealed tests, held out from all these choices and each scored once but for one recount after a frame-numbering error, are reported with the results. owl-5's final exam scores sealed tests 1 to 15 again, on the parts it did not train on and, of test 6, only the 83-frame thermopile set (Section 14.7).

Six real benchmark sets. Six test splits no model trained on: the Lepton 3.5, the camera Flux uses, in indoor rooms (SDT, 2,000 frames) and a crowded classroom and lab (IISER, 1,375 frames, 4,882 labelled people), a ceiling camera (2,347 frames, 8,043 people), rooms with hot laptops (265 frames), wall-mounted meeting rooms (813 frames, 1,061 people) and a side view (2,151 frames, 3,872 people). A frame is *exact* when the count equals the truth and *within one* when it is off by at most one person. The rule is scored as deployed and with its settings tuned on each set's validation split. Intervals are 95% block bootstraps, which recompute the score on 2,000 random redraws of whole runs of consecutive frames within a scene or sequence, because neighbouring frames are near copies: 50 SDT frames within a pose block, 25 IISER frames within a room, 100 PUT frames within a sequence, 20 SenSys frame numbers within a recording regime, 25 AAU annotated frames within a scene, and OpenThermalPose2 files grouped by subject and session prefix, with bare-numeric file names in runs of 50 (40, 56, 25, 14, 34 and 49 blocks). They describe sampling variation within these datasets, not across unseen venues.

Two properties of the sets matter. IISER's labels mark visible faces, so a person facing away is absent from its truth (a recount is in Section 14.3). And some splits share scenes: AAU's three meeting rooms and their people appear in training and test, IISER's test frames share their two rooms with training, and one PUT test sequence continues a training sequence. Removing that sequence and every test frame nearly identical to a training frame raises Owl's PUT score from 92.9% to 96.0% exact and moves no other set by more than 0.1 points, so near-copies do not inflate the results; AAU was not re-split, so its figure comes from rooms and people Owl trained on.

Stress test. 41 hard situations, each generated on purpose 150 times and scored alone, counted exactly as the sensor counts, crowd model included. Each case's random seed is fixed, so a rerun draws the same frames. The cases come from the same generator as 40% of the training data and guided which cases training emphasised, so the stress test checks coverage of known hard cases; the real sets carry the headline numbers. Its intervals are Wilson intervals for a proportion, which treat the frames as independent.

Blind-spot report. Every real benchmark set split by situation (by pose for SDT and by number of people elsewhere), and every stress case. Anything under 90% exact or 97% within one is flagged.

14 Owl: results

Figure 14 and Table 17 compare the pair Flux ran until 3 November, owl-4.9 and its crowd model, with the rule on all six real benchmark sets. On five of them Owl counts 82–96% of frames exactly and is within one person on 96–100%, where the deployed rule manages 9–77% exact. On the crowded classroom set, the hardest of the six, it is exact on 39% and within one on 74%, against the deployed rule's 33% exact; here the two are statistically close on exact counts (a margin of 6.3 points, interval −5.5 to 17.6), and Owl's lead grows with crowd size.

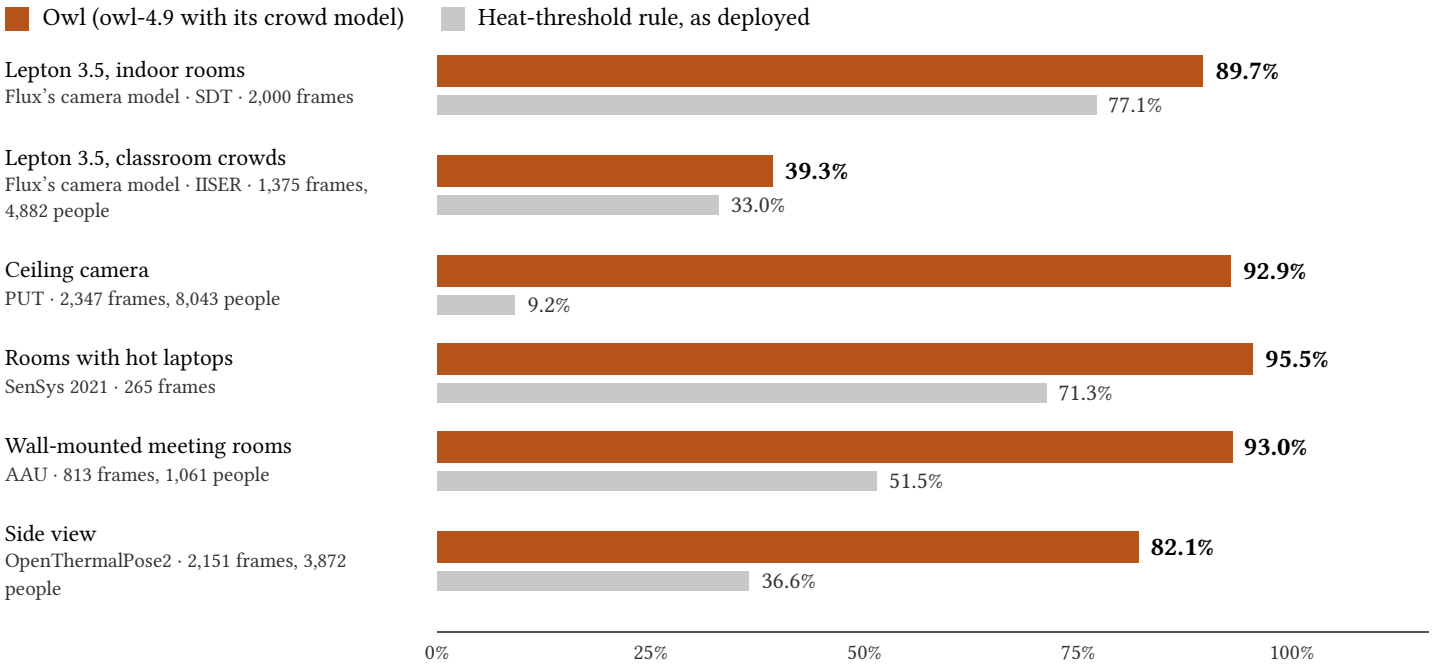


Figure 14. Frames counted exactly right on six real benchmark sets that no model trained on, owl-4.9 with its crowd model against the heat-threshold rule as deployed. The rule is scored on identical frames; Table 17 adds intervals and the rule tuned to each set.

Table 17. owl-4.9 with its crowd model against the heat-threshold rule on identical frames: the six real benchmark sets, the sixteen sealed tests and the stress test. Percent of frames, with 95% intervals for Owl's exact rate: block bootstrap on the benchmarks; frame-level Wilson intervals on the sealed tests and the stress mean, which ignore dependence within a recording (the margins in the text resample blocks). The last column is the rule with its cell size, warmth margin and minimum group size tuned on each benchmark's validation split; no sealed set was tuned.

Benchmark set	Owl exact	Owl within one	Rule exact, as deployed	Rule exact, tuned per set
Lepton 3.5, indoor rooms (SDT)	89.7 (85.4–93.4)	100	77.1	93.7
Lepton 3.5, classroom crowds (IISER)	39.3 (31.2–47.7)	73.9	33.0	40.2
Ceiling camera (PUT)	92.9 (86.9–97.2)	99.7	9.2	29.4
Rooms with hot laptops (SenSys 2021)	95.5 (91.7–98.2)	100	71.3	82.6
Wall-mounted meeting rooms (AAU)	93.0 (86.4–98.0)	99.1	51.5	75.5
Side view (OpenThermalPose2)	82.1 (74.4–89.2)	96.3	36.6	60.6
Sealed test: office, 1–2 people (OpenThermalAction)	94.3 (91.1–96.4)	100	40.7	–
Sealed test 2: another 32 × 24 sensor (SKKU)	33.7 (24.7–44.0)	86.5	79.8	–
Sealed test 2: outdoor handheld camera, people and dogs	41.0 (33.8–48.6)	86.1	16.9	–
Sealed test 3: 32 × 32 ceiling thermopile, 1–5 people (Klir)	9.2 (5.1–16.1)	32.1	18.3	–
Sealed test 4: lab room, 3 views (BU-TIV Lab)	38.0 (33.3–42.9)	81.8	7.3	–
Sealed test 4: atrium, 2 views (BU-TIV Atrium)	26.2 (23.1–29.7)	65.0	35.7	–
Sealed test 4: sports hall, 0–7 players (AAU soccer)	5.1 (3.3–7.7)	25.2	10.4	–
Sealed test 5: 12 public sets, pooled (Roboflow Universe)	40.4 (38.7–42.2)	78.4	33.0	–
Sealed test 6: 3 public sets, pooled (no account needed)	69.7 (67.0–72.3)	90.4	52.2	–
Sealed test 7: industrial hall, groups of 1–7 (UPPET)	8.8 (5.7–13.4)	29.8	20.5	–
Sealed test 8: a café and a restaurant, pooled	5.1 (1.4–16.9)	20.5	5.1	–
Sealed test 9: factory floor, hot robot arms (WLRI-HRC)	22.7 (19.0–26.9)	59.0	3.0	–
Sealed test 10: harbour front, outdoors (LTD)	16.7 (12.3–22.4)	44.0	3.8	–
Sealed test 11: room of 1–5 walkers, head height (LITIV)	3.1 (0.9–10.7)	28.1	18.8	–
Sealed test 12: 32 × 24 ceiling sensor, 0–5 people (MlDetection)	19.0 (15.7–22.8)	55.3	39.7	–
Sealed test 13: living rooms and offices, warm furniture (Thermal-IM)	64.0 (58.0–69.6)	90.8	54.4	–
Sealed test 14: 32 × 24 doorway sensor (TIDOS)	10.2 (7.9–13.1)	31.0	34.1	–
Sealed test 15: one person, 3 views (Almería)	88.6 (86.0–90.7)	99.7	44.9	–
Sealed test 16: railway platforms (VIPER)	26.9 (23.6–30.4)	74.0	17.4	–
41-case stress test, mean	94.7 (94.1–95.2)	99.3	–	–

A tuned rule. Owl, one model for all six sets, leads the rule tuned on each set's validation split by 13 to 64 points on four of them. On the two Lepton sets the tuned rule is level with or ahead of Owl: 93.7% against 89.7% on SDT, whose frames hold one person or none and where most of Owl's errors are missed people (149 of 206), and 40.2% against 39.3% exact (84.6% against 73.9% within one) on IISER. Adding the sensor's learned background mask made the tuned rule worse on every video set, because people who sit still fade into it.

A sealed test. To measure Owl on footage that guided nothing, I set aside 300 frames from the office test split of OpenThermalAction, a public MIT-licensed thermal set that no Owl version, setting or threshold had seen, spread across 269 clips of one or two people, at least 30 frames apart within a clip. Owl's and the rule's counts were stored first. The frames were then counted visually in two passes of one vision-language model (an AI that reads images), each blind to both outputs and to the other, with a third pass wherever they disagreed; the passes agreed on 93.9% of the 734 frames counted this way (these 300 and the 434 classroom frames recounted below), and on the sealed set the counts match the dataset's design on 298 of 300 frames. No person has checked them yet. Scored once, Owl led by 53.7 points (clip-block bootstrap 47.3 to 60.0), 94.3% exact (95% interval 91.1 to 96.4) to the deployed rule's 40.7%, and was within one on every frame to the rule's 88.7%.

Beyond its training cameras. A second sealed test asked how far Owl travels, with truth from each set's own person boxes (duplicates merged). On 89 frames from another 32 × 24 sensor (SKKU MLX90640, MIT licence), Owl trailed 33.7% to 79.8% exact: it took close, blurred bodies for screens. On 166 outdoor photos from a handheld Seek camera (Thermal Dogs and People, CC BY 4.0), Owl led 41.0% to 16.9%, but on the 70 frames showing only a dog it counted a person in 59%. A Voxel51 set, then the only one with many frames of three or more people, was scored and withdrawn for an unconfirmed licence; Owl trailed the rule on it. A third sealed test used the one licence-clean public set then found with indoor groups of three or more, 109 unseen frames from a 32 × 32 ceiling thermopile (Klir et al., GPL-3.0): the rule was exact on 18.3% against Owl's 9.2%, and led 24.7% to 3.9% on the 77 frames of three to five people, as Owl undercounts the faint, blurred bodies of a sensor it never trained on.

Crowded rooms and a sports hall. A fourth sealed test added the two public sets of larger groups then obtainable. BU-TIV [35], used for evaluation with its authors' permission and, from November, for training (Section 14.7), films an indoor lab and an atrium with 16-bit cameras at 512 × 512. On 395 Lab frames from three views, most holding three to five people, Owl counted 38.0% exactly (33.3–42.9) and 81.8% within one; the rule, 7.3% and 29.1%. On the 338 Lab frames of three or more people Owl led 34.6% to 3.6% exact, a margin of 31.1 points (block bootstrap 25.1 to 37.8). In the Atrium, 686 frames of mostly lone, distant people, the rule led 35.7% to 26.2%: a threshold rarely misses one distant warm body, which Owl, trained on closer views, sometimes drops; only Owl got empty frames right (32.0% against 0%). Across 1,081 BU-TIV frames Owl led by 5.2 points (0.4 to 10.1), and by 16.3 on frames of three or more. On 393 soccer crops filmed from high in a sports hall [36], players 25 to 50 pixels wide in 640-pixel-wide crops (6 to 12 in Owl's input), both failed (Owl 5.1% exact, the rule 10.4%).

Twelve more public sets. A fifth sealed test used the thermal person sets on Roboflow Universe whose licence permits this use and whose labels say which box is a person: 12 sets under CC BY 4.0 or a public-domain dedication, 2,976 unseen frames after one frame per source image, near-duplicate removal and the unseen check. Six other downloads were set aside before any prediction: two re-uploads of non-commercial datasets, one traced to the source of the withdrawn Voxel51 set, and three whose classes do not say which label is a person. Pooled, Owl counted 40.4% exactly (38.7–42.2) against the rule's 33.0%, a margin of 7.4 points (block bootstrap 3.2 to 11.6), and was within one person on 78.4% against 64.9%. In an exploratory split by sensor, Owl led by 10.0 points (5.0 to 14.5) on the nine sets from camera-resolution sensors and trailed by 6.5 (−14.7 to 1.4) on the three from 32 × 24 arrays; an occupancy set from a 32 × 32 array, reported apart because it comes from the group whose Lepton set Owl trained on, went the same way (6.6% against 70.0%). On the 222 pooled frames of three or more people the rule led, 20.3% to 9.9%.

Four more, with no account needed. A sixth sealed test used sets anyone can download under CC BY 4.0. On 530 frames from a 720 × 576 camera, almost all of one person, Owl led 90.2% to 45.8% exact. On 519 frames from libraries, cafés, gyms and outdoor scenes at 256 × 192 (TPD, whose labels can miss a bystander), Owl trailed by 4.8 points in exact agreement, 58.2% against 63.0% (block interval −12.8 to 2.9), and on its 23 frames of three or more labelled people the rule led, 34.8% to 4.3%. On 83 frames from a 32 × 32 thermopile, Owl and the rule showed no clear difference (10.8% against 25.3% exact; interval −38.1 to 3.5). Pooled over these 1,132 frames, Owl led by 17.5 points (92 blocks, seed 20261014: 8.5 to 26.1); on the 84 pooled frames of three or more people the rule led, 17.9% to 4.8%. On 79 outdoor FLC-Rural frames, reported apart, Owl and the rule showed no clear difference (22.8% against 11.4% exact; interval −3.6 to 19.4). The unseen check removed one set before any score was read: 474 of its 486 candidate frames matched frames Owl was trained or validated on (32 × 24 correlation 0.95 or more), because it is SenSys 2021, an Owl training source, under another title.

A hall, a café, a restaurant, a factory floor and a harbour. Four more sealed tests ran on 15 October. UPPET's TPE split [37], evaluated with Fraunhofer IOSB's permission, never to be trained on, films staged groups of one to seven people from high in an industrial hall, 10 to 40 pixels tall in Owl's input; on 215 frames from 18 sequences (six sparsely labelled ones set aside before any prediction) both undercounted, Owl exact on 8.8% and the rule on 20.5% (margin −11.6, block interval −25.7 to 0.5). The eighth took the two public videos closest to a venue: in a café of 5 to 19 people neither counter was ever exact (Owl averaged 4.0 people against 8.9 present), and in a restaurant of one to six each was exact on 2 of 10 distinct frames, Owl within one on 8 and the rule on 6. On two held-out sessions of WLRI-HRC [38], a factory floor where one or two people work beside two hot robot arms, Owl led by 19.7 points (14.8 to 23.8), 22.7% exact against 3.0% on 432 frames; both overcounted (Owl on 74.1% of frames with one person or none, the rule on 99.5%), Owl's false counts falling mostly on ceiling lights and a warm machine, rarely on the arms. On 209 held-out frames of the outdoor LTD harbour front [40], Owl led by 12.9 points (7.8 to 18.1), 16.7% exact against 3.8%, on empty and sparse frames where the rule counted warm pavement and cars; on the 90 frames of three or more people, typically 9 pixels tall in Owl's input, the rule led, 8.9% to 1.1%.

A room of walkers. An eleventh sealed test, also on 15 October, used the LITIV 2014 set [41], by permission: one to five people walking, standing and passing each other in a 5 m room, filmed at head height and rectified to 480 × 360, with the authors' count of each annotated frame as truth. On 64 frames from the two held-out videos, 205 people, Owl trailed by 15.6 points (block interval −28.9 to −3.3), 3.1% exact to the rule's 18.8%. The rule counted every lone walker. Owl gave 76 person peaks for the 205 people, 75 of them on a person, and put 33 of its 68 screen detections on people. With three or more people both failed (rule 4.3%, Owl 2.2%).

A ceiling sensor in offices and homes. A twelfth sealed test, on 18 October, used MlDetection [43], by permission: a 32 × 24 MLX90640 on the ceilings of offices, homes and labs, pointing down or at 45°, with the authors' person boxes as truth and their six test clips, from rooms outside their training set, held out whole. On 479 frames (1,041 people), Owl trailed by 20.7 points (block interval −27.6 to −13.8), 19.0% exact to the rule's 39.7%, and by 17.9 on frames of three or more (21.6% against 39.4%). It counted 4.0 people a frame where 2.2 were present, and 8.5 in empty rooms: the sealed tests map each image's 5th to 99.5th percentile onto 20 to 34 °C, and that stretch turns a still room's small differences into warm spots. Given the sensor's own temperatures, as on Flux, Owl led by 12.1 points (6.1 to 18.2), 32.6% to 20.5%, in a second comparison also fixed before any prediction. The sensor maker's software (Melexis), whose output ships with the set, was exact on 49.8%.

Heat left on furniture. A thirteenth sealed test, on 20 October, used Thermal-IM [44] (BSD-3-Clause): one person in living rooms and offices, filmed a second after touching or sitting on furniture that keeps the heat. One frame from each held-out clip (261) was counted visually in two blind passes of a vision-language model, which agreed on every frame and have not yet been checked by people. Owl counted 64.0% exactly against the rule's 54.4% (block interval of the difference −13.7 to 34.3). On frames with one person it saw a second on 25.8% (the rule 44.5%), mostly on chairs and cushions still warm from that person, and both counted someone in all five empty rooms whose chair was still warm: a weakness the generated stress test, where warm footprints score 98%, does not show. Thermal-IM's training clips did not train Owl. Its only person labels are body poses in the colour camera's image; one mapping onto the thermal image put the head on the head in about 21 of 40 checked frames, short of the 36 set in advance, and no frame could be confirmed empty.

People passing under a doorway sensor. A fourteenth sealed test, on 20 October, used TIDOS [45], by permission: a 32 × 24 MLX90640, not Flux's 160 × 120 Lepton, looking down from above two classroom doors. Its labels count the room, not the view, so its 519 frames were counted visually in two blind passes of a vision-language model (agreeing on 516; not yet checked by people). Owl trailed by 23.9 points (block interval −29.5 to −18.7), 10.2% exact to the rule's 34.1%, finding people in 96% of empty frames, where the stretch made noise into warm spots. On the sensor's own temperatures it got 95% of empty frames right but split bodies that fill the view, exact on 62.2% to the rule's 67.2%.

One person at the Lepton's size. A fifteenth sealed test, on 21 October, used the University of Almería's multi-view set [46], by permission: one person posing in a furnished room, filmed by three wall devices whose 160 × 120 thermal frames hold no temperatures. Its 699 test frames, each showing one person, were counted visually in two blind passes of a vision-language model (agreeing on all; not yet checked by people). Owl led by 43.6 points (block interval 38.6 to 48.9), 88.6% exact to the rule's 44.9%; of its 80 errors, 55 were a second person at one of two fixed spots where nobody stood.

Platforms far beyond a room. A sixteenth sealed test, on 22 October, used VIPER [47], whose owners allow scientific use only and allowed its results on this website: 651 frames of railway platforms with 0 to 4 people, outdoors, with the authors' labels as truth. Owl led by 9.5 points (block interval 1.0 to 18.2; −1.2 to 17.4 without five doubtful label segments), 26.9% exact to the rule's 17.4%, but neither count rose with the people in view.

14.1 From owl-1 to owl-4.9

Owl went through 16 named versions in four days; Figure 15 follows those shipped on the five original real sets (Findings 1, 2 and 9).

Owl, exact ---- the rule

Missing points were not measured for that version.

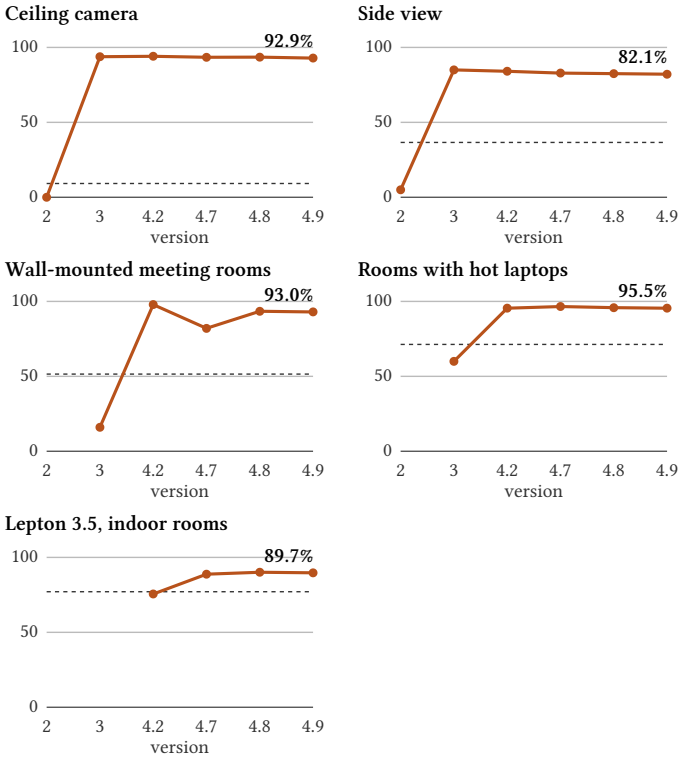


Figure 15. Exact counts on each real benchmark set's test split by version. owl-4.2 was the first shipped with meeting rooms, laptop rooms and measured physics; owl-4.7 and owl-4.8 added frames from the Lepton 3.5; owl-4.9 added real classroom crowds. owl-3's side-view figure includes the background fix of Finding 3 (74% before it).

14.2 Like-for-like comparison of owl-4.8 and owl-4.9

After training owl-4.9 I re-graded it and owl-4.8 on the full test splits with identical code (Table 18). On the five earlier sets no exact rate moved over 0.6 points and no within-one rate over 0.8.

Table 18. owl-4.8 and owl-4.9, each with its crowd model, graded together on full test splits. Exact / within one, in percent. The owl-4.8 stress mean comes from a run before each case's seed was fixed; owl-4.9's is the seeded rerun.

Test set	Rule exact	owl-4.8	owl-4.9
Lepton 3.5, indoor rooms	77.1	90.1 / 100	89.7 / 100
Lepton 3.5, classroom crowds	33.0	11.5 / 20.6	**39.3 / 73.9**
Ceiling camera	9.2	93.5 / 99.7	92.9 / 99.7
Rooms with hot laptops	71.3	95.8 / 100	95.5 / 100
Wall-mounted meeting rooms	51.5	93.4 / 99.9	93.0 / 99.1
Side view	36.6	82.5 / 96.6	82.1 / 96.3
41-case stress test, mean exact	–	94.5	94.7

14.3 Crowds on the Lepton 3.5

IISER's classroom set is my only real crowd from the Lepton 3.5: rows of 12 to 15 students, up to ten with a visible face, each row partly hidden behind the one in front. owl-4.8 had seen nothing like it and overcounted, averaging 8.1 people against 3.6 labelled faces. Figure 16 separates the two halves of the fix. Training the main model on classroom frames lifted exact counts from 11.5% to 28.6%. Retraining the crowd model as well took them to 39.3%, and on frames with six to ten labelled people moved within-one from 3.4% to 96.1%.

Owl's strength is the crowd: on frames with six to ten labelled people it is within one on 96.1%, against the deployed rule's 4.5%. Sparse classroom frames are the opposite case. Against the face labels, the rule is exact on 37.9% and 72.7% of frames labelled with one or two people, against Owl's 14.6% and 22.4%, and on the 88 frames labelled empty Owl averages 2.6 people. Every classroom frame labelled empty or with one face falls in IISER's test split (training holds 17 two-face classroom frames and none with fewer), so Owl never trained on an empty or one-person classroom. The 334 test frames labelled with two or fewer faces, and 100 others, were recounted visually, every person with any part in view, in the sealed test's blind two-pass procedure: the face labels undercount (frames labelled empty averaged 1.1 people, and none was empty), which narrows Owl's error on sparse frames. On frames holding one or two people the rule leads, 85% exact against Owl's 21% to 22%, partly because Owl puts points on warm seat backs; on frames holding three or more, Owl leads, 43.5% to 21.1% exact. In an exploratory check that mixes the two label definitions, with the 434 recounted frames in place of their face labels, the IISER test reads 41.5% exact and 77.7% within one for Owl, against 40.4% and 69.2% for the rule.

When the crowd model steps in is a single setting, chosen on the classroom frames and the stress test: handing over earlier was worse on both (Table 19), so the sensor keeps five.

Table 19. When to ask the crowd model, owl-4.9 pair. The stress means and blind-spot counts come from runs before each case's seed was fixed, which can differ by up to 10 points between draws; the seeded rerun of the shipped setting gives a stress mean of 94.7%.

Crowd model asked from	Classroom exact	Within one	Stress mean	Blind spots
3 people	36.9	72.3	94.3	12
4 people	37.5	73.1	94.6	10
5 people (shipped)	**39.3**	**73.9**	**94.9**	11

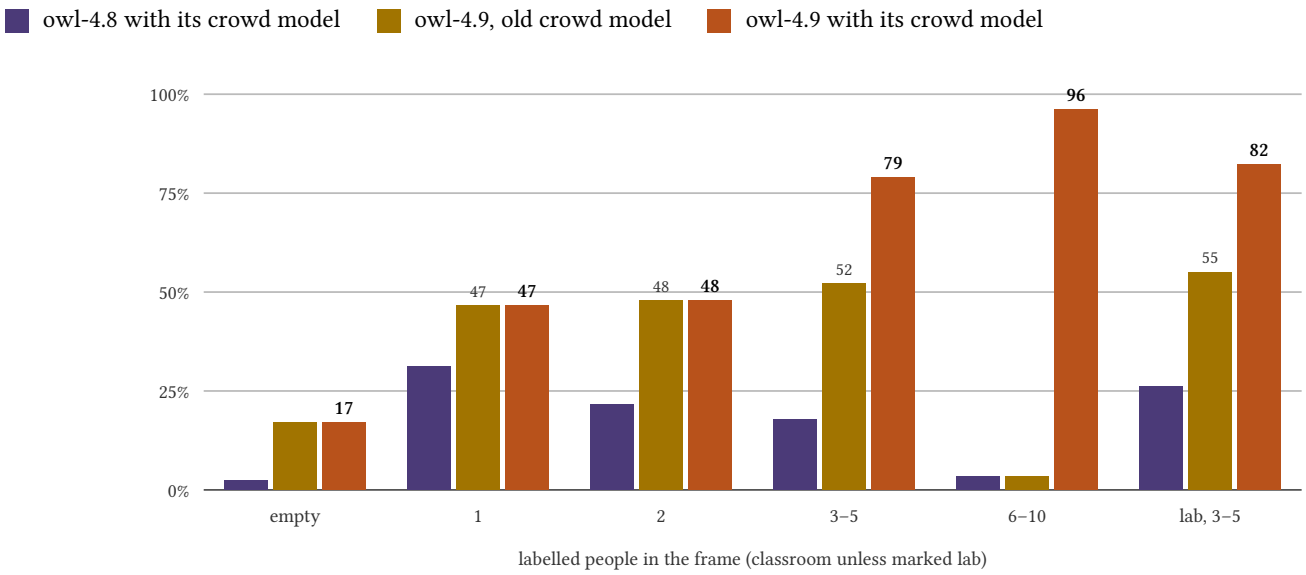


Figure 16. Frames within one person of the truth on IISER's held-out classroom and lab frames, by how many people were labelled in the frame. The crowd model's contribution is the difference between the gold and orange bars; it decides the six-to-ten bucket almost on its own.

14.4 Hard cases

Figure 17 shows all 41 stress cases for owl-4.9. In generated scenes, every warm object that fools the rule by design is counted correctly on 98 to 100% of stress frames, where the deployed rule manages 14% to 85%: a hand at the lens, a hot mug, a monitor, a radiator, a pet, a reflection, warm footprints, a candle; real heat left on furniture is harder (sealed test 13). Four cases fall under 90% exact (Figure 17): people lying flat, a bare torso at the lens, one person behind another (99.3% within one) and a crowd of ten (31.3%, 95% interval 24.5 to 39.1; 75.3% within one).

14.5 Naming everything else

On 3,000 held-out generated frames, owl-4.9 finds the nine other kinds as Table 20 shows. These names are shown on the sensor's screen to explain why something was not counted.

Table 20. Naming on held-out generated frames. Found: share of real objects with a peak. Right when named: share of named peaks of the right kind.

Kind	Objects	Found	Right when named
person	10,461	74.9%	89.7%
hand	682	69.5%	84.3%
pet	617	67.1%	92.4%
hot drink	836	84.2%	83.0%
food	441	81.2%	91.6%
laptop	768	68.2%	79.8%
screen	491	60.5%	80.7%
heater	794	78.7%	85.5%
lamp	316	77.5%	79.0%
warm seat	586	58.7%	83.1%

The same 3,000 frames are deliberately harder than real rooms, often with crowds of up to ten. On them owl-4.9 leads the rule 59.3% to 20.2% exact and 84.3% to 49.9% within one; empty rooms are right on 92.3% and single people on 80.1%. The generator grew harder with each version, so these figures do not compare across them.

14.6 Speed and size

On one core of the sensor's Pi 5, the main model of owl-4.8 and owl-4.9 takes 14–16 ms per frame to run and decode, measured by the sensor's self test (ten reads) on its camera. A frame with five or more people also runs the crowd model, a network of the same size (722,655 exported weights against 717,422 for the main model, whose 718,550 training parameters shrink slightly at export), which roughly doubles that frame's time; that path has not yet been timed on the Pi. owl-5 keeps both networks' design and size; it has not been timed on the Pi.

14.7 owl-5, the November retrain

owl-5 is owl-4.9 retrained with the training parts of the sets that tested it, wherever their terms and labels allow: 25 sets for candidate A and 22 for candidate B (Table 21). The split, fixed on 15 October before any training and extended on 19 and 20 October with the earlier rows unchanged, holds 184,576 frames: 61,831 for training, 8,140 for validation and 114,605 held out, each version hashed. Each set is cut along its own unit: BU-TIV by camera view, MlDetection by room, WLRI-HRC by session, LTD by day, LITIV 2014 by video, Klir's sensor by take and OpenThermalAction by scene; the fifteen Roboflow sets, the café and restaurant among them, and the soccer match by time (the first 70% of each Roboflow take trains or validates; the match's last sequence is held out); SKKU and the dogs set by capture number. No frame both trains and grades: the 1,842 training and validation frames within a 32 × 24 correlation of 0.95 of a held-out frame left training, 13 of them already out as failed reads, and no training frame came that close to a benchmark test frame (largest 0.895). The sealed tests owl-5 meets again were all scored with owl-4.9 before it trained, so for owl-5 they are reused evaluation sets; its untouched test is 215 frames of LITIV 2012 [42], counted blind before any model read them.

BU-TIV's first author, who granted its evaluation on 15 October, allowed training on it and a trained model in the product on 24 October, provided the original data are never distributed for commercial purposes; as with every set here, I show none of its frames and cite it [35]. Three of its five camera views train owl-5: 17,950 training frames, plus 3,032 validation frames in five clips of the same views. The other two views, 12,402 frames, are never trained on; the final exam scores 491 of them, on sealed test 4's grid.

MlDetection's authors and Melexis allowed training with citation on 19 October [43]: its ten training rooms train, its two validation rooms validate, and its six test clips, from rooms it never saw, stay held out.

Table 21. What owl-5 adds to owl-4.9's training data (Table 16, all kept), as packed for training, with the sealed test that scored each set's held-out part in brackets (for OpenThermalAction, sealed test 1's office split). A label "x / y" gives the training and validation units. People are counted in the training frames. PD: public domain. A: candidate A only.

Set (test)	Train / val frames	People	Terms
BU-TIV, 3 of 5 views [35] (4)	17,950 / 3,032	62,065	Permission
MlDetection, 10 / 2 rooms [43] (12)	18,312 / 824	38,649	Permission
WLRI-HRC, 5 / 2 sessions [38] (9)	2,034 / 514	2,191	CC BY 4.0
SKKU MLX90640 (2)	2,211 / 396	2,442	MIT
Thermal Dogs and People (2)	118 / 22	81	CC BY 4.0
15 Roboflow sets, first 70% (5, 8)	5,151 / 938	7,726	CC BY 4.0, PD, MIT
LTD, 42 / 6 days [40] (10)	386 / 78	1,631	CC BY 4.0
LITIV 2014, video 2 [41] (11)	46 / 0	95	Permission
OpenThermalAction, balcony (1)	7,074 / 882	14,148	MIT, AGPL labels; A
Klir HTPA 32 × 32 (3)	4,261 / 754	4,326	GPL-3.0; A
AAU soccer, sequences 1–3 [36] (4)	2,323 / 429	10,056	CC BY-SA 4.0; A
Added, A	59,866 / 7,869	143,410	
Added, B	46,208 / 5,804	114,880	

Two candidates. Three sets carry share-alike terms that could be read to require a model trained on them to be shared on the same terms: Klir's frames (GPL-3.0), the soccer hall (CC BY-SA 4.0) and OpenThermalAction's pose labels (tagged AGPL-3.0). So two candidates trained from one recipe: A on every trainable set, B on the same sets without those three. The rule between them was fixed on 2 November, before either trained. It compares their exact counts on identical validation frames of the sets both trained on, in two pools: all 12,597 such frames, and the 5,554 with three or more people. A counts as clearly better only if, on either pool, it leads by at least 2.0 points with the lower end of its 95% block-bootstrap interval above zero; otherwise B goes forward. The bootstrap resamples each set's coarsest independent unit: a session, camera view, room, scene or subject where the data show separate sessions, and the whole set where they do not (SenSys 2021, SDT, SKKU, Thermal Dogs and People, LTD and 14 of the 15 Roboflow sets), 49 units in all and 17 among the frames of three or more. The share-alike question was for me to decide: on 2 November, before the comparison, licensing Owl to other companies became part of Flock's plan, so A would go forward only if clearly better, and then only if I chose it.

Recipe. Both candidates repeat owl-4.9's last training run with the new data added: 20,000 steps from owl-4.9's weights at a peak rate of 2×10^{-4} and batches of 64, each frame drawn from the generator with probability 0.4 and otherwise from the real pool. That pool is shared in proportion to training frames, no new set above a quarter of it and owl-4.9's sources splitting their part in their old ratios (MlDetection and BU-TIV, the largest, drew 18.6% and 18.2% of A's real frames and 21.6% and 21.2% of B's). Each crowd model trains the same way from owl-4.9's crowd model. MlDetection trains in degrees Celsius and the other new sets on their sealed tests' fixed mapping (5th percentile to 20 °C, 99.5th to 34 °C), their head points spread as for the body-box sets. The generator's warm-seat option and dining scenes stayed off. Each candidate chose its crowd model (its own or owl-4.9's), its threshold (0.35, 0.40 or 0.45) and its crowd hand-over (four, five or six people) on validation frames alone. The 27 files that training and scoring rest on, the training packs, split, scripts and starting weights among them, were hashed beforehand, and the runs started only after every hash matched.

Final exam. The gate was fixed on 15 October, before any training, and its frame grid on 16 October. owl-5 replaces owl-4.9 only if all three hold: (1) pooled over the held-out parts of BU-TIV, the soccer hall, SKKU, Thermal Dogs and People and Klir's sensor (1,593 frames), its exact count beats owl-4.9's, with a block-bootstrap interval above zero; (2) on the 333 of those frames with three or more people it gains at least 5 points; (3) on each of the six benchmark test splits and on sealed test 1 it loses no more than 2.0 points. The frozen candidate takes the exam once, through software that refuses a second candidate or a changed file, with sealed test 4's statistics. Beside the gate, deciding nothing, it is scored on the held-out parts of sealed tests 5 and 7 to 11, on sealed tests 12 to 15 and on the 83-frame thermopile set of sealed test 6, the radiometric ones both with the sealed tests' mapping and in degrees Celsius; sealed test 13, whose failures motivated the unused warm-seat option, is labelled so.

Result. Candidate B, frozen on 3 November, was exact on 52.8% of the new sets' validation frames (26.0% with three or more people), where owl-4.9 managed 36.3% (14.0%), and on 82.7% of owl-4.9's own six validation parts, against 83.5%. It asks its own crowd model from four people and uses a threshold of 0.35. On the 12,597 validation frames both candidates share, A minus B was −0.4 points of exact count (−1.4 to 1.1), and −0.1 (−2.3 to 4.1) on the 5,554 with three or more people, so A was not clearly better on either pool, and B went forward.

In its final exam (Table 22) owl-5 passed the first two criteria. Pooled over the five gate sets it was exact on 37.2% of 1,593 frames against owl-4.9's 31.5%, 5.8 points (0.7 to 12.0), and on their 333 frames of three or more people it gained 9.3 points (28.5% against 19.2%). It failed the third, losing more than 2.0 points on three benchmark sets, PUT (−3.1), OpenThermalPose2 (−3.4) and IISER (−6.0), so the exam's verdict is a fail, and the refit planned for a pass was not trained. Candidate B never trained on the soccer hall or Klir's sensor, which supply 988 of the 1,593 frames and 96 of the 333 crowded ones, and BU-TIV's 491 frames come from two camera views it never saw, of rooms it trained on.

Table 22. owl-5's final exam against owl-4.9 on identical frames: exact count in percent, owl-5 minus owl-4.9 in points (from unrounded rates) and its 95% block-bootstrap interval (whole sequences for LITIV 2012). Every set but LITIV 2012 was scored with owl-4.9 before owl-5 trained. Mapped: the sealed tests' fixed mapping; °C: the sensor's own temperatures, as Flux reads them. Test 13's failures motivated the warm-seat option, which stayed off. Two of MlDetection's frames fail the read-fault rule its training used and stay, as the test's frames are fixed.

Set (frames)	owl-4.9	owl-5	Change	95% interval
The gate, fixed 15 October				
1. All five sets (1,593)	31.5	37.2	+5.8	0.7 to 12.0
2. 3+ people (333)	19.2	28.5	+9.3	2.7 to 17.1
BU-TIV, 2 views (491)	29.9	54.2	+24.2	16.1 to 32.0
Soccer, sequence 4 (108)	12.0	18.5	+6.5	−10.2 to 25.0
SKKU 32 × 24 (52)	46.2	76.9	+30.8	14.8 to 41.3
Dogs and people (62)	45.2	41.9	−3.2	−19.7 to 13.1
Klir 32 × 32 (880)	32.8	27.4	−5.5	−10.0 to −0.2
3. No set more than 2.0 points down				
SDT (2,000)	89.7	88.4	−1.3	−2.9 to 0.2
IISER (1,375)	39.3	33.3	−6.0	−12.4 to 0.3
PUT (2,347)	92.9	89.9	−3.1	−8.1 to 0.5
SenSys 2021 (265)	95.5	96.2	+0.8	−1.8 to 4.1
AAU (813)	93.0	98.2	+5.2	0.0 to 14.3
OpenThermalPose2 (2,151)	82.1	78.6	−3.4	−5.1 to −2.1
Sealed test 1 (300)	94.3	93.0	−1.3	−3.6 to 0.7
Beside the gate				
Test 5, held out (897)	35.7	57.9	+22.2	17.0 to 27.6

Set (frames)	owl-4.9	owl-5	Change	95% interval
Test 5, IISER set (146)	2.7	45.2	+42.5	35.7 to 51.2
Test 6, mapped (83)	10.8	8.4	−2.4	−11.7 to 7.0
Test 6, °C (83)	8.4	6.0	−2.4	−9.4 to 9.4
Test 7, UPPET hall (215)	8.8	18.6	+9.8	−0.4 to 20.8
Test 8, held out (14)	0.0	0.0	0.0	0.0 to 0.0
Test 9, factory (432)	22.7	80.6	+57.9	51.2 to 63.9
Test 10, harbour (209)	16.7	17.7	+1.0	−6.9 to 8.4
Test 11, walkers (64)	3.1	62.5	+59.4	46.8 to 70.8
Test 12, mapped (479)	19.0	28.6	+9.6	3.0 to 16.2
Test 12, °C (479)	32.6	33.8	+1.3	−3.9 to 6.9
Test 13, Thermal-IM (261)	64.0	67.4	+3.4	−2.4 to 10.1
Test 14, mapped (519)	10.2	34.1	+23.9	18.7 to 29.3
Test 14, °C (519)	62.2	61.5	−0.8	−4.6 to 2.8
Test 15, Almería (699)	88.6	95.4	+6.9	3.3 to 10.3
Untouched				
LITIV 2012 (215)	38.1	57.2	+19.1	−2.1 to 38.0

Beside the gate, owl-5 led owl-4.9 with an interval above zero on both parts of sealed test 5, the factory floor, the walkers, MlDetection mapped, TIDOS mapped and the Almería set; on the rest the interval includes zero (Table 22).

The untouched test. On the 215 LITIV 2012 frames owl-5 was exact on 57.2%, owl-4.9 on 38.1% and the rule on 29.8%. Resampling the nine sequences whole, owl-5 minus the rule was 27.4 points (19.8 to 34.9), owl-4.9 minus the rule 8.4 (−8.3 to 26.4) and owl-5 minus owl-4.9 19.1 (−2.1 to 38.0). On its 27 frames of three or more people the three were exact on 44.4%, 11.1% and 25.9%. The 205 frames on which both blind counts agreed give the same reading.

On Flux. owl-5 counts the scenes owl-4.9 never saw, and their crowds, better: the gate's pool and crowded frames above, and the held-out parts of sealed tests 5, 9 and 11 by 22 to 59 points; on the untouched frames it was exact on 57.2%, owl-4.9 on 38.1%. So on 3 November I put it on Flux in owl-4.9's place, departing from the plan, under which a fail keeps owl-4.9. Its cost is the three benchmark losses above; only OpenThermalPose2's interval excludes zero. It ships exactly as scored, without a refit.

Limits. Cuts by time or capture number leave neighbouring frames on both sides of a cut (only those at a correlation of 0.95 or more left training), and BU-TIV's validation clips are interleaved with its training clips. LITIV 2012's counts are model-assisted and not yet checked by people.

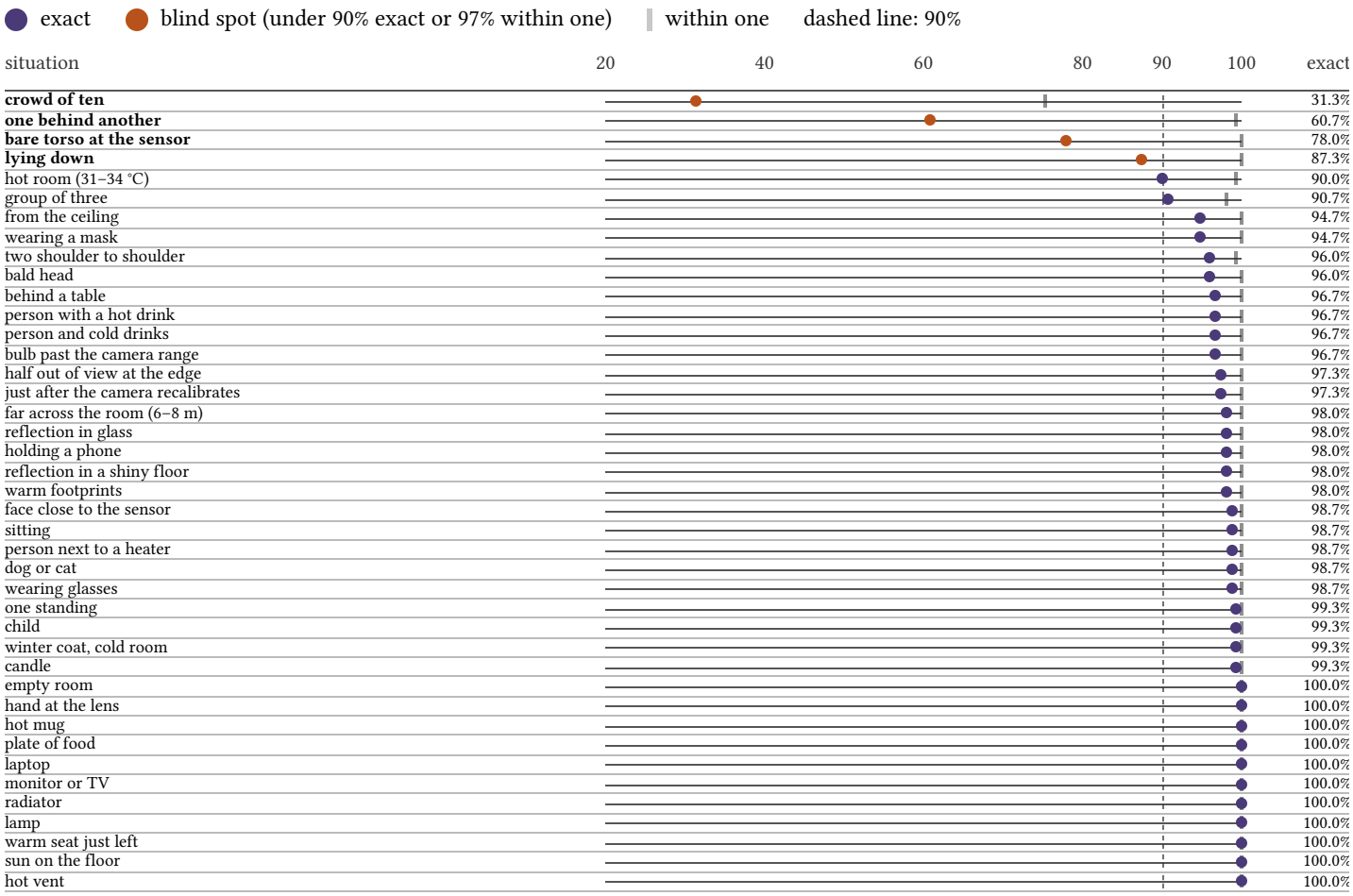


Figure 17. The 41-case stress test for owl-4.9 with its crowd model, 150 generated frames per case from a seed fixed per case, sorted from hardest to easiest.

15 Owl: findings

The real benchmark sets decided every change to Owl. These results shaped it, in the order I found them, including the alternatives set aside. Stress-test figures here come from runs before each case's seed was fixed and can differ by up to 10 points between draws.

F1 ***Generated frames alone fail on real footage***

owl-2 counted 80% of generated frames exactly and found 73–94% of the warm objects, but counted 0% of real ceiling frames and 5% of side-view frames exactly: it had only seen clean drawn shapes and read a real room's texture as people.

F2 ***A little real data fixes most of it***

A 500-step test with real frames mixed in went from 0% to 81% exact on the real ceiling set. The full run, owl-3, reached 94% on the ceiling set and 74% on the side view. From then on, generated frames also gained real-looking texture and low-resolution blocking.

F3 ***Judge warmth against the cool end of the frame***

The physical check compared each person with the room, taken as the frame's median; when a person fills the view the median is the person, so the check threw them out. Taking the room as the 10th percentile lifted the real side view from 74% to 85% exact and 86% to 98% within one, hot rooms from 47% to 68%, and a bare torso at the sensor from 16% to 30%, with the ceiling set and every false-alarm case unchanged.

F4 ***Physics beats tricks***

Two training changes that seemed sensible, skewing the absolute-temperature channel and drawing real group frames three times as often, changed nothing on the ceiling and side-view sets and cost a few points on unseen rooms. Replacing hand-picked temperatures with measured skin and clothing physics lifted the unseen meeting rooms from 24% to 33% exact and from 58% to 85% within one.

F5 ***A bigger network did not help***

Widening the network from 24 to 32 base channels tied the smaller one (94.2% and 73.9% against 93.8% and 74.2% on the ceiling and side view). Accuracy was limited by data, so later versions added data rather than width.

F6 ***More of a hard case can cost more than it gains***

owl-4.4 drew more people lying down and bare bodies at the sensor, the two single-person cases still under 90%. The real meeting rooms fell from 97.9% to 60.9% exact with no measurable gain on those cases, so the change was reverted.

F7 ***A sharp target at a guessed point teaches the model to miss***

The September 80 × 60 trial, a finer output grid, keeps heads a few pixels apart as two people. But on the datasets whose head points are estimated from a body box, a sharp target at the guess taught it to miss people: the meeting rooms fell from 97.9% to 67.2%. Spreading those targets over the pixels the head could occupy fixed the cause, and became standard for every later version.

F8 ***Density counts crowds that peaks merge***

Used as a second look in crowds, the finer-grid model lifted a crowd of ten from 41% to 63% within one. Back on the 40 × 30 grid, a density head, whose total does not fall when heads touch, lifted it to 82% and put one person behind another at 100% within one, with the four real benchmark sets then in use unchanged. It became the crowd model (owl-4.3).

F9 ***Data shares trade off, and the right data ends the trade***

Adding real Lepton frames (SDT) helped on that camera, but every way of making room hurt the meeting rooms: cutting their share to 15% cost 25 points, and owl-4.6 gained 13 points on the Lepton while losing 14 on the meeting rooms. owl-4.7 chose the camera Flux uses (88.8% against owl-4.2's 75.6%). Adding SDT's rendered Lepton-style frames in owl-4.8 improved both at once (Figure 18).

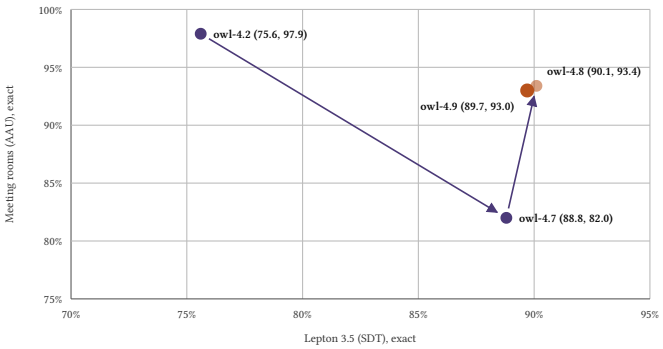


Figure 18. The trade between the Lepton 3.5 frames (SDT, horizontal) and the wall-mounted meeting rooms (AAU, vertical). owl-4.7 bought the camera with the meeting rooms; owl-4.8, trained with rendered Lepton frames as well, recovered the meeting rooms without giving the camera back. owl-4.9 differs from owl-4.8 by under one point on both sets.

F10 ***Validation scores do not compare across versions***

When training drew group frames three times as often, validation reused that weighting by accident, and owl-3.2 looked worse than owl-3 on a harder mix. Validation follows each run's mix of sources, so a validation score picks a checkpoint within its run and is never compared across versions.

F11 ***Lower precision cost more than it saved***

An 8-bit export ran 2.1 times faster at a quarter of the size, but cut owl-3's exact count on the real ceiling set from 94% to 86%, and to 84% when calibrated on real frames. At 14–16 ms per frame in full precision, speed was never the constraint, so the sensor keeps full precision.

F12 ***A newer base can make a worse crowd model***

Retraining the crowd model starting from owl-4.8 instead of its own earlier weights dropped a crowd of ten to 18.7% exact. The older crowd model stayed until real crowd data arrived.

F13 ***A real room shows what no test set does***

On the sensor in a real room, owl-3 named a nonexistent pet, sometimes counted a monitor, and missed a bare torso near the camera. Three changes answered them: higher naming bars for rarer kinds, dropping a person where a screen-like object is more confident, and bare skin and bodies filling the view in generated frames. On the stress test a bare torso at the sensor went from 16% under the first owl-3 decoder (30% after Finding 3) to 78% under owl-4.9 with fixed seeds.

F14 ***Real crowds on the Lepton 3.5 were the missing data***

After owl-4.8, training on the same data had stopped helping. Only after IISER Bhopal licensed its Lepton 3.5 classroom crowds to me did the crowd numbers move: 11.5% to 39.3% exact, an average error of 4.66 people to 1.03, and rooms of six to ten from 3.4% to 96.1% within one, while the other benchmark sets held steady (Section 14.2).

F15 ***Validation stopped improving early in the run***

owl-4.9's real validation score stopped rising about a quarter of the way through its run (Figure 19). The rest of the run brought no sustained gain, so the next experiment is new data: more real crowds and side views from this camera.

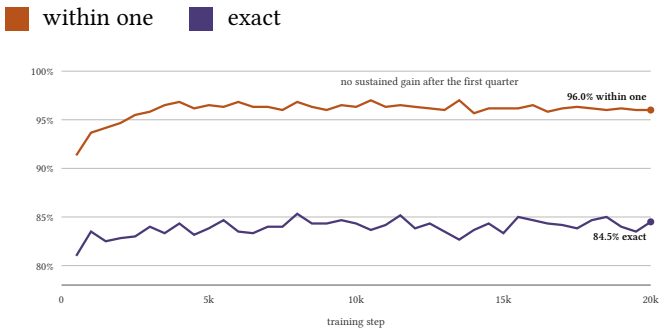


Figure 19. owl-4.9's real validation score every 500 steps over its 20,000-step run. Both curves are flat after about 5,000 steps. The validation mix includes the new classroom crowds, so these values are lower than earlier versions' and are not comparable with them.

16 Privacy by design

Flux is built to tell people how busy a place is. It must never become a camera that watches them.

- **Kinds, never who.** Owl's outputs are ten kinds of thing. It has no identity output, no face class and no face features. IISER labels faces, and I use each face box only as the point where a head is; Owl is trained to locate people, never to predict who they are.
- **No memory between frames.** Owl reads each frame on its own and tracks no one. Flux's doorway sensor follows a crossing only as long as it takes to tell an arrival from a departure.
- **Numbers are all that leave.** The network's points and boxes exist only in memory. Boxes are drawn only on units with a screen and are never sent. What leaves the sensor is a handful of numbers: people counts and a sound level, never an image.
- **Enforced by tests.** Tests in the sensor's suite fail the build if Owl gains an identity output or carries anything between frames, or if an image, a point or a box leaves the device.
- **Data used as licensed.** Every dataset is used as its licence or permission allows. None of their images are redistributed or published, and none are shipped with the model. IISER's licence also forbids using its data to identify anyone.

The model files are private, like Flock's forecasting model. The generator, network and training and grading code are public. Without a model on disk the sensor counts with the rule, and says so.

17 Future work on the counter

The clearest next step is data: owl-5 added training data from the sealed tests' own sets (Section 14.7), and the gap that matters most is real radiometric crowds from the Lepton 3.5 in venues, measured against hand counts. Flux is built to carry a doorway counter, an 8 × 8 time-of-flight sensor that tells arrivals from departures; it first ran on the bench in October 2026 and has not yet counted a real doorway. Flux's software already combines its running total with Owl's count of people in view and estimates how long people stay as occupancy divided by the arrival rate (Little's law [28]); tested in a real doorway, that would answer how many people are inside as well as how many are visible, without identifying anyone. Finally, each crowd decision currently rests on one frame; agreeing a count across consecutive frames, without following individuals, may steady crowded readings further.

18 Conclusions on the counter

The second question asked whether a credit-card-sized computer model can be designed to count the number of people inside a venue using a thermal camera without identifying individuals. Yes in simple rooms, not yet in crowds, and not yet inside a working venue. On one core of a Raspberry Pi, at 14 to 16 ms a frame, Owl counts people on a 160 × 120 thermal camera and sends only counts, with no identity output and no memory between frames (Section 16). owl-4.9 is 82 to 96% exact on five of six development benchmarks (rule 9 to 77%) and 94.3% on a sealed office set (rule 40.7%), and leads the rule when 15 further public sets are pooled in two sealed tests, on a factory floor beside hot robot arms and on an outdoor harbour front. It falls short on low-resolution arrays, people walking past a head-height camera, distant people on railway platforms and dense groups in a sports hall, an industrial hall or a café; owl-5 trained on several of these, but not the share-alike sports hall and Klir's sensor, the evaluation-only TIDOS, industrial hall and platforms, or test 6's thermopile. owl-5 counts the scenes owl-4.9 never saw better and runs on Flux from 3 November by my choice, though it failed the gate fixed before it trained (lost 3 to 6 points on 3 benchmarks; Section 14.7).

19 How it was built

Both studies were built in eight months, from the first collector in March 2026 to Owl's retrain in November. The forecast's spring-trained model, rebuilt after an August audit, gave way in September to a correction built on live readings; Owl replaced the heat rule in four days and was retrained in November (Section 14.7). Dates are Eastern time.

19.1 The forecast: corpus and first models (March to May 2026)

The first collector, committed on 10 March 2026, widened to 34 cities that day. From the start it gathered two kinds of row from the busyness provider: weekly rows, one typical week per venue of 168 hourly values on the provider's 0 to 100 scale, and realtime rows, one reading per venue-hour on a real date. Event listings were attached that week, and a model pipeline with an ONNX inference service followed on 14 March. In early May the collector reached the 13 of its 34 cities not yet tried: a probe of 100 venues in each found 996 of 1,300 (76.6%), and those entered the corpus. Beijing (37 of 100 found) stayed thin and holds no rows in v2.6.0's training data, so 30 cities train the model and three are held out (Section 4.5). The last spring reading is dated 18 May 2026, when the corpus held 3,912,357 rows across 34,784 venues in 34 cities (Section 4.2).

The first models predicted busyness directly, with the venue's popular-times curve among their inputs. One such 80-feature model scored far higher on validation than on held-out cities: it had learned to copy the curve, and coordinates acted as a lookup table for its five training cities. That result set three design decisions that still stand. Coordinates were dropped as features. Cross-validation holds out a whole city at a time. And the label became the *delta*, how far the hour departs from the venue's own curve, with the served score the curve plus a clamped correction. On 10 May, v2.2.1 read the curve from the server's table, the first fix making training and serving agree (train/serve parity). No accuracy figure from before August is used as evidence: the corpus was later repaired (Section 4.2).

19.2 An end-to-end audit, and v2.6.0 (August 2026)

In August I audited the whole pipeline and found two silent failures. The model had never loaded in production, because the package that reads its ONNX file was missing from the server, so the rule engine had answered every request. The collector had written no rows since 19 May while reporting success, because its provenance check compared zero rows written with zero rows labelled. Both now fail loudly. On 12 August, v2.3 trained on the rows production serves, realtime rows with a usable curve, because weekly rows carry a delta

label of zero, which had taught earlier models to shrink every correction. v2.3 to v2.4 failed their gates, and v2.5.0 shipped that afternoon. The next six days went to the data and the pipeline:

- *Train/serve parity* (14 August). Serving did not build training's inputs: live predictions used coordinates 0, 0, serving read all of a venue's place types where training read the first three, and the event-size estimator had drifted from the one that labelled the corpus. A parity break now stops the model loading.
- *The clock* (15 August). The weekly collector stored the provider's slot number as the hour; the provider's day runs from 06:00, so slot 18 was local midnight. A database fix corrected 3,536,360 rows; a row with no declared clock is now refused.
- *A second audit* (15 August). A read-only pre-retrain audit listed 21 findings, 8 of them blocking, among them 491,100 duplicate rows, an impossible month on 62.9% of rows and a gate that scored rows production never serves. All eight were fixed before the next run.
- *Provenance* (15 August). The 457,402 spring realtime rows cannot be split into live readings and provider forecasts; two attempts failed against null controls. Those rows stay marked unknown, and since 4 September training excludes them by default.
- *A category leak* (16 August). Two category features had been fitted on every city, so a held-out city helped build its own scores. They are now refitted without it.

The first retrain with every stage checked shipped as v2.6.0-starling on 18 August (Table 6). The gate's old within-10 floor of 29.2% predated the clock fix, so it was replaced by a measurement, the incumbent's 19.3% on Table 6's rows, which v2.6.0 beat at 20.7%. Of two copies, trained on a CPU and a GPU and within 0.02 MAE of each other, the CPU copy shipped because its run reproduces bit for bit.

19.3 Alternatives tested, and the decisions they produced (August to September 2026)

After v2.6.0, every proposed change faced a gate before it could ship, and each result is recorded so the experiment is not repeated.

- *Calendar features* (19 August). Varying the month alone moved v2.6.0's raw correction by 27.31 points on average, against 5.92 for the hour: with one spring of data, calendar features had absorbed collection drift. v2.7, without month and season, failed its gate (MAE 29.89, within-10 19.9%), so v2.6.0 stayed; the November retrain swaps month features for live readings (Section 7.2).
- *Calibration* (20 August). About 40 output corrections were fitted on the earliest 30% of dates and scored on 46,101 later rows. There a quantile display map gained 8.40 points of within-10 while raising MAE by 3.15. It went live on 28 August; September's live readings later showed it cost within-one accuracy, and it left the served number from 26 September to 1 November (Section 6.6).
- *Hour ranking* (20 August). Within one venue-night the model ordered two hours correctly 62.7% of the time and the curve 63.1%, over 31,498 pairs, so the app's "go later" advice reads the curve and stays silent below a 10-point gap.
- *Event provenance* (20 August). 22 of 34 cities had no row with a nearby event: no events were collected there. A provenance column now tells a measured absence from a missing lookup.
- *Game nights* (30 to 31 August). Games of seven teams became six features. Game nights are 7.1 points busier in the labels, but the spring training data held about 19 days of games, and the features did not move within-10. They stay switched off until the retrain.
- *The two-head model* (5 September). It had separate heads for weekly and live rows. Held to the verified live labels then available, it lost to the shipped model, and a model given 304,000 more rows of the same kind lost to one trained without them in a single paired run (Section 6.6). Both results pointed to the venue's own recent readings as the input to build.
- *Forty-two proposals* (6 September). Six research rounds proposed 42 changes; an automated reviewer, told to refute each one, left seven standing. The one shipped was the per-venue trailing offset (Section 6.6).
- *The curve's range* (6 to 8 September). The poor calibration of Section 3.1 invited a slope correction; it lowered MAE but cost within-10, so the system keeps the per-venue offset (Section 6.6). The same week measured what coverage is worth (Section 6.5), which made adding venues with a curve the collector's first job.

19.4 Live collection (September 2026)

Live collection restarted on 1 September for about 1,400 venues in Philadelphia and the Lehigh Valley, and every row now carries a provenance label. Calls settled at one a second after the provider's rate limit throttled the first day's 600 a minute. The sweep went hourly from 20:08 on 3 September (00:08 UTC on the 4th) and now runs at seven minutes past each hour. An open-hours filter skips venues their own curve says are shut, cutting the mean sweep from 1,414 calls to 993. From 5 September, Miami, one of the three held-out cities, is read at two fixed hours a day for the gate. From 6 September the venues users open lead every sweep, and from 25 September calls run in parallel, up to eight at once, so the full sweep finishes inside the hour. The pool of verified live readings grew from 4,223 on 4 September to 9,339 over 1,228 venues on 6 September.

19.5 The switch to curve, offset and nowcast (25 September to 1 November 2026)

On 25 September a replay of the serving code (the band replay) over every live reading in the 8 September export first measured what the card is scored on: whether the served number lands in the same crowd level as the live reading. As served, v2.6.0 with the display map landed within one level 57.4% of the time on 7,920 readings, and a reading one hour old predicted the next within one level 92.0% of the time. Eleven causes of error were ranked by the points each could recover, and the two largest needed no retrain. Two serving switches followed: the curve plus 0.85 of the trailing offset, and the nowcast of Equation (2), with weights fitted on 1 to 5 September and scored on 6 to 8 September. The switches went through six rounds of adversarial review by automated tools, one of which caught the offset seeing the hour it was scoring; the fix restricted it to readings strictly before the hour. They went live at 18:55 on 26 September, on the three-day result in the last row of Table 4. On 25 and 26 September the retrain's split rule and thresholds were committed (Section 7.2), and on 26 September the retrain was rehearsed (Section 6.7). On 8 October a fresh read-only export scored the switches on every live reading since 9 September (Section 6.1). On 13 October, when persistence baselines showed a two-hour reading no longer helped, production switched to rule (b), the simpler rule the retrain gate compares against (Sections 6.1 and 6.3). On 1 November it switched to model mode: v2.6.0-starling with the display map, offset and nowcast of Section 5.2.

19.6 The parity tests

The train/serve parity faults of Section 19.2 were the hardest to see, so the safeguards built since are tests that training and serving agree:

- the ONNX export verifies feature order, and serving refuses a model file listing a feature it cannot build;
- before any fold is fitted, a run with no city held out must reproduce the shipped category features bit for bit, and a control that rebuilds them the old, leaky way must be caught;
- the curve smoothing, the neighbour features (on 8,006 readings), the live features, the game features and 62 calendar, weather and

event columns are each computed by trainer and server on shared test data, with zero disagreements allowed;
- the collector's event values match serving's on 400 listings;
- the band replay must match serving's output on every row;
- a leak test changes every reading at or after a row's hour, its own label included, and none of the row's inputs may move;
- the serving responses of v2.6.0 must match recorded answers under every switch combination.

19.7 Flux, the sensor (May to October 2026)

The sensor began as backend tables and an endpoint on 2 May 2026, tested end to end before any device existed. The demo unit was designed on 4 May, at about $1,000 of parts: a Raspberry Pi 5, a FLIR Lepton 3.5 on a PureThermal 3 board, a 4G board, an analog microphone, an infrared break-beam and a battery. The Lepton was chosen over the 32 × 24 MLX90640 for its 160 × 120 temperature image (19,200 pixels against 768), and the sensor program moved onto it on 26 August. The first bench day on a Pi 5 was 6 September: the camera came up first time, and pooling the image into 4 × 4 cells stopped a bare head and a clothed torso reading as separate people. On 21 September the enclosure was designed in OpenSCAD and the device split in two, a base with the Pi, battery and screen, and a wall head whose sensors look down at a doorway. On 7 October an I2C decibel meter replaced the analog microphone; it measures A-weighted sound level on its own processor, so the Pi will read a number, never audio. On 8 October the base and wall head became one box, because the doorway sensor and the decibel meter need short I2C wires.

19.8 Owl, the training chain (September to November 2026)

Owl's versions, runs and findings are in Sections 12.5, 14.1, 14.7 and 15. Licence requests went out on 27 September; TU Wien granted SDT on 28 September, and IISER Bhopal licensed its Lepton 3.5 classroom crowds on 29 September. A dataset whose authors had not replied was not used, and another, OpenThermalAction, was left out of owl-4.9 for its copyleft labels. owl-4.9 and its crowd model, shipped on 29 September, draw 60% of their training frames from licensed datasets (SDT's rendered Lepton frames among them) and 40% from the generator. For owl-5 the split was fixed on 15 October and extended on 19 and 20 October, BU-TIV's first author allowed training on 24 October, and on 2 November the rule between candidates A and B, the recipe and 27 file hashes were fixed and the four runs trained, from 21:46 to 23:40. Candidate B was frozen on 3 November, failed its exam and went on Flux that day by my choice.

19.9 The doorway counter (September and October 2026)

The doorway counter went through three designs. The May plan's infrared break-beam became a one-sided infrared proximity module on 21 September, with one part to mount and nothing to align. On 25 September it became a VL53L8CX time-of-flight sensor that looks down at the doorway as an 8 × 8 grid of distances and tells arrivals from departures. On 26 September, before the part arrived, the manufacturer's driver was ported to the Python standard library. On 7 October the counter first ran on the bench, reading 265 frames.

19.10 Collecting the data, and asking for it (September and October 2026)

The forecast's readings come from the provider's API through Flock's hourly collector (Section 19.4), stored with their provenance and never redistributed. Everything else came from searching for public data and asking for the rest. A survey of open occupancy data found no public set that pairs a busyness score with counts of people, so the replication of Section 6.8 and its extensions use counts alone: 32 sources and 6,530,784 stored hourly records, credited in the acknowledgements. The 15 core datasets were analysed under a plan hashed before their first download; the 17 extensions are exploratory, and of 247 imported campus Wi-Fi access points, 153 qualified for scoring.

Where a licence was missing or unclear, I asked. On 14 and 15 October I sent more than 40 requests: to city transport and footfall teams (Wellington, York, Toronto, Amsterdam), university groups (KIT Library, Munich, UCL's GeoDS, Unisport Bern, Kassel), pool, gym and venue operators (Zurich, SWM Munich, Fitnesspark, The Cliffs), occupancy companies (Occuspace, hystreet, TouringPlans, NickOccupancy) and the owners of 26 thermal datasets, among them BU-TIV, TIDOS and two Lepton sets, one filmed in Prague's National Museum. KIT Library granted its Seatfinder counts, through its API and with credit, and the St. Gallen and Geneva university libraries their own. BU-TIV's first author granted evaluation of the Atrium and Lab sequences within a day (sealed test 4) and, on 24 October, training (Section 14.7). UPPET's authors allowed evaluation, not training (sealed test 7), WLRI-HRC's authors confirmed its CC BY 4.0 licence for evaluation and training (sealed test 9), the LITIV lab at Polytechnique Montréal granted evaluation and training (sealed test 11), KU Leuven allowed academic use of MlDetection (sealed test 12; with Melexis, training on its training rooms from 19 October) and VIPER (sealed test 16; its licence allows scientific use only, and on 30 October KU Leuven allowed its results on this website), Boston University research use of TIDOS (sealed test 14), and the University of Almería evaluation and training on its multi-view set (sealed test 15). Fitnesspark declined, so its counts, and a public archive of them, were left out. The rest are pending.

The same rules held for every source. A set with no licence and no permission is not used (IPHD). A portal refusing scripted requests is asked by email, not worked around (York). A set whose uploader licensed material it did not own is withdrawn, even after scoring (the Voxel51 thermal set). No test-set frame is published. Where a set's labels are faces (IISER), each face box serves only as a head's position.

20 Limitations

The first four limit the first answer; the last two limit the second.

- **Relative readings, not headcounts.** The main forecast study's labels are the provider's relative readings on a per-venue 0 to 100 scale, estimates from its own system. The forecast therefore predicts the provider's next live value, and a score of 70 means different crowds at different venues. No count inside a venue has checked these readings; their weekly rhythm tracks subway entries beside 695 New York venues (median r 0.74, Section 6.8). On 15 public real-count datasets the provider-specific findings do not hold as stated, so they describe the provider's estimates rather than crowds in general. But 13 of those datasets count people passing a sensor; the two registered occupancy datasets and the KIT, Singapore and Barcelona series added later side with the provider: the hour-old reading beats the curve (Section 6.8).
- **The test window.** The served accuracy rests on 119,939 readings over 30 dates in Philadelphia, the Lehigh Valley and Miami, three of the 34 cities in one season; the nowcast's gain comes from the two Pennsylvania regions alone. The online correction shipped on a three-day test, short of the five dates the band gate requires; the 30-day test confirms it. Rule (b) was identified on this window (Section 6.1) and has not yet been scored on dates not used to choose it; the retrain's November window will supply them (Section 7.2). When the served number says *Packed* it is right far more often than its predecessor (67.4% against 28.0%), but catches fewer Packed hours (45.3% against 49.2%, on 20,445 readings).

- **The delta model's data and holdout.** Section 6.2 rests on spring rows of unknown provenance, and its three held-out cities are also the validation set. On verified live readings the spring-trained model loses to the weekly pattern: MAE 30.94 as served (display map on) against 26.40 for popular times over the 30-day test, and 27.52 against the curve's 26.42 for the model without the display map or offset, on 7,920 readings in the 8 September export. Production has served this model since 1 November, with the display map, offset and nowcast of Section 5.2; this paper does not score that configuration. Ten of its 106 features are never used (Section 5.1), and latitude, dropped as a feature, re-enters through daylight hours. Table 12's learner comparison is untuned and confined to three cities.
- **A silent feed.** From 12 to 23 October the provider returned almost no live readings. Over 28 replayed outage start dates, losing the hour-old reading costs 20.7 points of exact level at once, then about 0.4 points a week for three weeks; an offset kept for a month would still beat the bare curve by 3.6 points, but each reading expires after 28 days, and rule (b) then falls back to the curve. Until a fix deployed on 15 October, the serving path also dropped offsets whose stored row was over seven days old.
- **Owl in the field.** Owl has not been measured in a working venue (Section 21), and its six real sets also served as development benchmarks (Section 13). The first sealed test (Section 14) is one or two people against a plain wall, from the lab whose side-view set Owl trained on, and its counts, like the IISER recounts, came from an AI model reading the frames, not from counting in the room. Owl transfers unevenly to cameras it was not trained on. The rule beat owl-4.9 on 32 × 24 and 32 × 32 arrays (sealed tests 2, 3, 5, 12 and 14; on test 12 Owl led when given the sensor's real temperatures), in BU-TIV's atrium of lone, distant people, in a room of walkers (test 11) and on pooled frames of three or more people (tests 5 and 6), and dense or distant groups defeated both counters (tests 4, 7, 8, 10 and 16). owl-5 trained on the training parts of the sets behind seven of these tests (2, 4, 5, 8, 10, 11 and 12; of test 4, BU-TIV only), so its results on their held-out parts (Section 14.7) are on sets it learned from, on the same cameras apart from BU-TIV's two unseen views and MlDetection's unseen rooms; where it never trained (the soccer hall of test 4, Klir's sensor of test 3, the office of sealed test 1, test 6's thermopile, tests 7 and 13 to 15, and LITIV 2012), it led owl-4.9 on TIDOS mapped and the Almería set, trailed it on Klir's sensor and showed no clear difference on the other eight. owl-5, which Flux runs, gave up 3.1 points on PUT, 3.4 on OpenThermalPose2 and 6.0 on IISER; the recording on Flux's own camera, the untouched test still to come, decides how it does in real rooms. Outdoors Owl counted a person in most frames showing only a dog. The hardest cases remain people hidden by people (a side view of three to five people is exact on 40% of frames, 85.5% within one), lying and sitting people on the Lepton (exact on 81% and 83% of SDT frames, which hold one person or none) and sparse classrooms, where Owl counts some warm seat backs.
- **No Flux is installed at a venue.** The one unit is a demonstration unit, Flock's crowd number reads no sensor data, and the doorway counter has run on the bench alone.

21 What comes next: the retrain, and joining the forecast to the counter

This section sets out what would strengthen each answer: the retrain for the first, a recording on Flux's own camera for the second, and counts from a working venue for both. Owl's retrain has run and is reported in Section 14.7.

21.1 The November retrain

Section 7.2 sets out the forecast's retrain. Owl's retrain, owl-5, is reported in Section 14.7: it failed the gate fixed on 15 October and runs on Flux from 3 November by my choice. UPPET (test 7), TIDOS (test 14) and VIPER (test 16) are never trained on. owl-5's successor, owl-5.1, will keep the new data, restore the old sets' share of training and be chosen on validation frames alone, then, as the exam cannot be reused, face a new untouched test before it replaces owl-5. The first is a recording on Flux's own camera, made with consent in a real room of one to fifteen people counted as it is filmed, kept unseen until owl-5.1 is final and scored once with owl-4.9, owl-5 and owl-5.1 together; sparse rooms, where Owl counts warm seat backs (Section 14.3), and dense groups are what it must answer first. LTDv2's labelled validation split, a newer, larger release of the LTD harbour set [40], is planned as a further test once a near-copy check against the LTD days owl-5 trained on clears it. The generator's warm-seat option, the heat a body leaves on a seat, cushion or table edge, stayed off for owl-5; sealed test 13's failures motivated it, so any gain from it will be judged on validation data and the Flux recording.

21.2 Counts as ground truth

The forecast predicts busyness normalised per venue, with 100 the busiest that venue gets (Section 4.1). A score of 70 could mean a dozen people in a small bar or two hundred in a large one. A Flux unit in a venue Flock already reads would pair its headcount with the busyness provider's live score hour by hour, showing how many people each score means there and whether the five crowd levels match what a group sees at the door. Its doorway counter, once proven on a real door, would add arrivals and departures, so the sensor could estimate how many are inside as well as in view (Section 17).

It would also give the served number a fresher reading: its strength is the latest reading, whose value sits in the first hour (Sections 3.2 and 6.3, Figure 8), and a sensor reporting every 30 seconds makes that reading seconds old, where readings now arrive at most hourly (Section 4.2). On real counts like Flux's, Section 6.8's replication favours adding the latest departure to the next hour's curve.

21.3 What would be measured first

One unit, installed with the owner's agreement at a venue the hourly collector already reads; hand counts of the people in Owl's view on several busy nights, to score Owl there before its counts serve as ground truth, and of arrivals and departures to score the doorway counter; and, hour by hour over several weeks, the busyness provider's live score against Flux's count, the score per person, and how often the crowd level matches what the count shows.

22 Conclusion

Can AI tell how crowded a venue will be before you go? Yes for the next hour, and only slightly further ahead. What does it is small: each venue's weekly curve corrected by its own recent live readings, which named the provider's crowd level one hour ahead 54.9% of the time, against 30.2% for the popular-times chart and 30.8% for the larger gradient-boosted model (Section 8). Those are the provider's estimates, not head counts. Can a credit-card-sized computer model be designed to count the number of people inside a venue using a thermal camera without identifying individuals? Yes in simple rooms, not yet in crowds, and not yet inside a working venue: on one Raspberry Pi core, sending only counts, Owl was exact on 94.3% of a sealed office set against the heat rule's 40.7%, and dense groups still defeat it (Section 18). owl-5, retrained in November, counts the scenes owl-4.9 never saw better and runs on Flux from 3 November by my

choice, though it failed the gate fixed before training (Section 14.7). The forecast's retrain and a first recording in a venue (Section 21) will test both answers again.

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Data sources. The public counts of Section 6.8 come from Brisbane City Council; the City of Melbourne Pedestrian Counting System; Dublin City Council and the National Transport Authority; Cork City Council; Dún Laoghaire-Rathdown County Council (via Smart Dublin); Leeds City Council (Data Mill North) and the City of York Council (York Open Data), both containing public sector information licensed under the Open Government Licence v3.0; the City of Toronto (contains information licensed under the Open Government Licence – Toronto); Heart of the City, Auckland; Kanton Basel-Stadt; Stadt Zürich (Open Data Zürich: the Bahnhofstrasse counts, measured by hystreet.com, and, under CC0, the Tiefbauamt's pedestrian counters and the Verkehrsbetriebe Zürich counts at Hardbrücke); the Cambridge Water Department; the Seattle Department of Transportation; the Metropolitan Transportation Authority (data.ny.gov); Crowdedness at the Campus Gym by Nick Rose (Kaggle, ODbL v1.0); and the KIT Library, Karlsruhe Institute of Technology, the University of St. Gallen Library and the University of Geneva Library, whose Seatfinder occupancy data are used with their permission. The open series of the sixth check come from Aarhus Kommune (opendata.dk, CC BY 4.0); Boryeong City (data.go.kr 15134612; no use restrictions); the University of Alberta Library (Borealis, doi:10.5683/SP3/TCY5XY, CC0); Saga City (BODIK, CC BY 4.0); Orléans Métropole (data.orleans-metropole.fr, Licence Ouverte v2.0, data of 15 October 2026); Montpellier Méditerranée Métropole (data.montpellier3m.fr, ODbL); and the Comune di Bologna (BolognaWifi Affollamento, CC BY 4.0). The room data come from ROBOD (Tekler, Ono, Peng, Zhan, Lasternas and Chong, National University of Singapore; figshare, CC BY 4.0). The campus Wi-Fi data come from Universitat Politècnica de Catalunya, Space and Time User Distribution in a University Campus (Mendeley Data, CC BY 4.0). The ceiling-sensor frames of the third sealed test come from Klir et al., Multi-Person Localization based on a Thermopile Array Sensor with Machine Learning and a Generative Data Model (GitHub, GPL-3.0). The fourth sealed test uses BU-TIV (Wu, Fuller, Theriault and Betke, Boston University), evaluated and, from November, trained on with its first author's permission and never redistributed, and the Thermal soccer dataset (Gade and Moeslund, Aalborg University; Kaggle, CC BY-SA 4.0). The fifth sealed test uses twelve sets from Roboflow Universe, each under CC BY 4.0 or a public-domain dedication and credited to its publisher: Thermal Human Detection; yolo-thermal-pedstrian-indoor; CIIS Thermal; Outcomex Thermal Person Detection; drone-work Thermal People; football-3arjv Thermal Human 4; mlx90640; thermalobjectdetection MLX90640 Camera Person Detection; Mondragon Unibertsitatea Thermal Camera Person Detection; pnusafetynet Thermal Human Detection; thermla-mlx90640; and toinonen Thermal Human; with IISER Bhopal's Occupancy Estimation (MIT) reported apart. The sixth uses the VILNIUS TECH thermal pose dataset (Vdoviak and Sledevic; Mendeley Data, CC BY 4.0), TPD (Dhillon, Chaturvedi, Sharma and Kumar, Bennett University; Zenodo, CC BY 4.0), Poudel's thermopile occupancy set (Zenodo, CC BY 4.0) and FLC-Rural (Dhillon, Sharma and Kumar; Zenodo, CC BY 4.0). The seventh to sixteenth use UPPET's TPE images (Cormier, Specker and Beyerer, Fraunhofer IOSB; CC BY-NC-SA 3.0, by permission), Thermal People Cafe (uon101) and Thermal Occupancy Counter (QuietBites Project) from Roboflow Universe, WLRI-HRC (Süme, Ponomarjova, Wendt and Rupitsch, University of Offenburg; Zenodo), the Long-term Thermal Drift dataset (Nikolov et al., Aalborg University; Kaggle), the LITIV 2014 thermal-visible dataset (Bilodeau, Torabi and colleagues, Polytechnique Montréal; by permission), MlDetection (Vandersteegen and colleagues, KU Leuven EAVISE with Melexis; by permission, with training on its training rooms allowed from 19 October), Thermal-IM (Tang, Ye, Ma and Zhao, Tsinghua University and MIT; BSD-3-Clause), TIDOS (Cokbas, Ishwar and Konrad, Boston University; by permission for research, never trained on, as commercial use of a model trained on it needs a Boston University licence), the Multi-View Visible-Thermal set (Lupión and colleagues, Universidad de Almería; Mendeley Data, CC BY-NC 3.0, by permission) and VIPER (Van Beeck, Van Engeland, Vennekens and Goedemé, KU Leuven EAVISE; for scientific use only, its results shown here by permission, never trained on); the untouched November test uses LITIV 2012 from Polytechnique (by permission). Except where noted, they are used under CC BY 4.0 or a public-domain dedication. The first sealed test uses OpenThermalAction (ISSAI, Nazarbayev University; frames under the MIT licence, pose labels tagged AGPL-3.0). owl-5 trained on the training parts of BU-TIV, MlDetection, WLRI-HRC, the SKKU set, Thermal Dogs and People, the fifteen Roboflow Universe sets of the fifth and eighth tests, the Long-term Thermal Drift dataset and LITIV 2014, and its candidate A also on those of Klir's set, the soccer dataset and OpenThermalAction (Section 14.7). Venue busyness readings come from a commercial venue-busyness provider, which has no part in and does not endorse this work. Weather data comes from OpenWeather (https://openweathermap.org/) under the Open Database License (ODbL). Sports schedules come from TheSportsDB and event listings from Ticketmaster. No dataset named here is released with this paper.

Every number behind Section 6.8 and Owl's tests is kept in Flock's research database, the sixteenth sealed test's in tables of its own, as its owners' terms require: the 6,530,784 hourly public counts with each source's licence and the attribution it requires; 65,780 frame-level count records (9,242 frames across fifteen rounds of sealed tests, 483 frames of the separately reported IISER occupancy set, 434 IISER recounts and the 215 LITIV 2012 frames, each with Owl's and the rule's count, owl-4.9's count on 9,924 validation frames from 27 sets, owl-5's on the 15,077 frames of its exam, side tests and LITIV 2012, and both owl-5 candidates' on their 14,662 validation frames), beside the owner, licence and terms of each of the 52 thermal datasets; the 184,576-frame November split (Section 14.7); and 1,820 result rows, each with the file that computes it.

References

[1] J. Bollenbach, S. Neubig, A. Hein, R. Keller, and H. Krcmar. 2022. Using machine learning to predict POI occupancy to reduce overcrowding. In *INFORMATIK 2022* (LNI P-326). Gesellschaft für Informatik, Bonn, 393–408. https://doi.org/10.18420/inf2022_34

[2] J. Bollenbach, S. Neubig, A. Hein, R. Keller, and H. Krcmar. 2024. Enabling active visitor management: local, short-term occupancy prediction at a touristic point of interest. *Information Technology & Tourism* 26, 3 (2024), 521–552. https://doi.org/10.1007/s40558-024-00291-2

[3] S. Chang, E. Pierson, P. W. Koh, J. Gerardin, B. Redbird, D. Grusky, and J. Leskovec. 2021. Mobility network models of COVID-19 explain inequities and inform reopening. *Nature* 589, 7840 (2021), 82–87. https://doi.org/10.1038/s41586-020-2923-3

[4] T. Chen and C. Guestrin. 2016. XGBoost: a scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*. ACM, 785–794. https://doi.org/10.1145/2939672.2939785

[5] J. B. Copas. 1983. Regression, prediction and shrinkage. *Journal of the Royal Statistical Society: Series B (Methodological)* 45, 3 (1983), 311–335. https://doi.org/10.1111/j.2517-6161.1983.tb01258.x

[6] F. Galton. 1886. Regression towards mediocrity in hereditary stature. *The Journal of the Anthropological Institute of Great Britain and Ireland* 15 (1886), 246–263. https://doi.org/10.2307/2841583

[7] A. Hajisafi, H. Lin, S. Shaham, H. Hu, M. D. Siampou, Y.-Y. Chiang, and C. Shahabi. 2023. Learning dynamic graphs from all contextual information for accurate point-of-interest visit forecasting. In *Proceedings of the 31st ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '23)*. ACM, 1–12. https://doi.org/10.1145/3589132.3625567

[8] R. J. Hyndman and A. B. Koehler. 2006. Another look at measures of forecast accuracy. *International Journal of Forecasting* 22, 4 (2006), 679–688. https://doi.org/10.1016/j.ijforecast.2006.03.001

[9] S. Islam, D. Gandhi, J. Elarde, T. Anderson, A. Roess, T. F. Leslie, H. Kavak, and A. Züfle. 2021. Spatiotemporal prediction of foot traffic. In *Proceedings of the 5th ACM SIGSPATIAL International Workshop on Location-based Recommendations, Geosocial Networks and Geoadvertising (LocalRec '21)*. ACM, 1–8. https://doi.org/10.1145/3486183.3490997

[10] S. Kapoor and A. Narayanan. 2023. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns* 4, 9 (2023), 100804. https://doi.org/10.1016/j.patter.2023.100804

[11] M. Möhring, B. Keller, R. Schmidt, and S. Dacko. 2021. Google Popular Times: towards a better understanding of tourist customer patronage behavior. *Tourism Review* 76, 3 (2021), 533–569. https://doi.org/10.1108/TR-10-2018-0152

[12] D. S. Moore, W. I. Notz, and M. A. Fligner. 2013. *The Basic Practice of Statistics* (6th ed.). W. H. Freeman and Company, New York.

[13] ONNX Runtime developers. ONNX Runtime. Software, accessed October 2026. https://onnxruntime.ai/. Server: onnxruntime-node 1.29.0; sensor: the Debian python3-onnxruntime package, version 1.21.

[14] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and H. Larochelle. 2021. Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program). *Journal of Machine Learning Research* 22, 164 (2021), 1–20.

[15] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, and C. F. Dormann. 2017. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* 40, 8 (2017), 913–929. https://doi.org/10.1111/ecog.02881

[16] S. Timokhin, M. Sadrani, and C. Antoniou. 2020. Predicting venue popularity using crowd-sourced and passive sensor data. *Smart Cities* 3, 3 (2020), 818–841. https://doi.org/10.3390/smartcities3030042

[17] X. Zhou, D. Wang, and P. Krähenbühl. 2019. Objects as points. arXiv:1904.07850. https://doi.org/10.48550/arXiv.1904.07850

[18] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár. 2017. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV '17)*. IEEE, 2999–3007. https://doi.org/10.1109/ICCV.2017.324

[19] V. Lempitsky and A. Zisserman. 2010. Learning to count objects in images. In *Advances in Neural Information Processing Systems 23 (NIPS 2010)*. 1324–1332.

[20] Y. Zhang, D. Zhou, S. Chen, S. Gao, and Y. Ma. 2016. Single-image crowd counting via multi-column convolutional neural network. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR '16)*. IEEE, 589–597. https://doi.org/10.1109/CVPR.2016.70

[21] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel. 2017. Domain randomization for transferring deep neural networks from simulation to the real world. In *Proceedings of the 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 23–30. https://doi.org/10.1109/IROS.2017.8202133

[22] C. Pramerdorfer, J. Strohmayer, and M. Kampel. 2020. SDT: a synthetic multi-modal dataset for person detection and pose classification. In *Proceedings of the 2020 IEEE International Conference on Image Processing (ICIP)*. IEEE, 1611–1615. https://doi.org/10.1109/ICIP40778.2020.9191284

[23] A. Samal and H. R. Lone. 2025. Thermal vision: pioneering non-invasive temperature tracking in congested spaces. *Smart Health* 36 (2025), 100576. https://doi.org/10.1016/j.smhl.2025.100576. Dataset: https://osf.io/9z5a4/

[24] C. Palmero, A. Clapés, C. Bahnsen, A. Møgelmose, T. B. Moeslund, and S. Escalera. 2016. Multi-modal RGB–depth–thermal human body segmentation. *International Journal of Computer Vision* 118, 2 (2016), 217–239. https://doi.org/10.1007/s11263-016-0901-x. Dataset: https://vap.aau.dk/

[25] P. Aszkowski and M. Piechocki. 2022. Thermo Presence: the low-resolution thermal image dataset and occupancy detection using edge devices. In *Proceedings of the 3rd Polish Conference on Artificial Intelligence (PP-RAI 2022)*. Publishing House of Gdynia Maritime University, Gdynia, 49–52. Dataset: https://github.com/PUTvision/thermo-presence

[26] A. Kuzdeuov, M. Zakaryanov, A. Tleuliyev, and H. A. Varol. 2025. OpenThermalPose2: extending the open-source annotated thermal human pose dataset with more data, subjects, and poses. *IEEE Transactions on Biometrics, Behavior, and Identity Science* 7, 4 (2025), 902–913. https://doi.org/10.1109/TBIOM.2025.3575499. Dataset: https://huggingface.co/datasets/issai/OpenThermalPose2

[27] S. Zhu, T. Voigt, D. F. Perez-Ramirez, and J. Eriksson. 2021. A low-resolution infrared thermal dataset and potential privacy-preserving applications. In *Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems (SenSys '21)*. ACM, 552–555. https://doi.org/10.1145/3485730.3493692. Dataset: https://doi.org/10.5281/zenodo.5574233

[28] J. D. C. Little. 1961. A proof for the queuing formula: $L = \lambda W$. *Operations Research* 9, 3 (1961), 383–387. https://doi.org/10.1287/opre.9.3.383

[29] Teledyne FLIR. FLIR Lepton 3.5 radiometric thermal camera module. Product documentation, accessed October 2026.

[30] K. He, X. Zhang, S. Ren, and J. Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR '16)*. IEEE, 770–778. https://doi.org/10.1109/CVPR.2016.90

[31] F. Yu and V. Koltun. 2016. Multi-scale context aggregation by dilated convolutions. In *International Conference on Learning Representations (ICLR 2016)*. arXiv:1511.07122.

[32] H. Law and J. Deng. 2018. CornerNet: detecting objects as paired keypoints. In *Computer Vision – ECCV 2018* (Lecture Notes in Computer Science). Springer, Cham, 765–781. https://doi.org/10.1007/978-3-030-01264-9_45

[33] Y. Li, X. Zhang, and D. Chen. 2018. CSRNet: dilated convolutional neural networks for understanding the highly congested scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR '18)*. IEEE, 1091–1100. https://doi.org/10.1109/CVPR.2018.00120

[34] M. Piechocki, M. Kraft, T. Pajchrowski, P. Aszkowski, and D. Pieczyński. 2022. Efficient people counting in thermal images: the benchmark of resource-constrained hardware. *IEEE Access* 10 (2022), 124835–124847. https://doi.org/10.1109/ACCESS.2022.3225233

[35] Z. Wu, N. Fuller, D. Theriault, and M. Betke. 2014. A thermal infrared video benchmark for visual analysis. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW '14)*. IEEE, 201–208. https://doi.org/10.1109/CVPRW.2014.39. BU-TIV dataset, by permission.

[36] R. Gade and T. B. Moeslund. 2018. Constrained multi-target tracking for team sports activities. *IPSJ Transactions on Computer Vision and Applications* 10 (2018), 2. https://doi.org/10.1186/s41074-017-0038-z

[37] M. Cormier, A. Specker, and J. Beyerer. 2025. UPPET: unified pedestrian pose estimation in thermal imaging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW '25)*. IEEE, 4551–4560. https://doi.org/10.1109/CVPRW67362.2025.00440

[38] S. Süme, K.-M. Ponomarjova, T. M. Wendt, and S. J. Rupitsch. 2025. Enhancing human–robot collaboration with thermal images and deep neural networks: the unique thermal industrial dataset WLRI-HRC and evaluation of convolutional neural networks. *Journal of Sensors and Sensor Systems* 14, 1 (2025), 37–46. https://doi.org/10.5194/jsss-14-37-2025

[39] D. Läufer, S. Braun, S. Süme, and U. Himmelsbach. 2024. Comparison of approaches for human detection with low-resolution infrared data sets using deep learning. In *2024 21st International Conference on Ubiquitous Robots (UR)*. IEEE, 596–602. https://doi.org/10.1109/UR61395.2024.10597494

[40] I. Nikolov, M. P. Philipsen, J. Liu, J. V. Dueholm, A. S. Johansen, K. Nasrollahi, and T. B. Moeslund. 2021. Seasons in drift: a long term thermal imaging dataset for studying concept drift. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, Vol. 1 (NeurIPS Datasets and Benchmarks 2021).

[41] G.-A. Bilodeau, A. Torabi, P.-L. St-Charles, and D. Riahi. 2014. Thermal–visible registration of human silhouettes: a similarity measure performance evaluation. *Infrared Physics & Technology* 64 (2014), 79–86. https://doi.org/10.1016/j.infrared.2014.02.005

[42] A. Torabi, G. Massé, and G.-A. Bilodeau. 2012. An iterative integrated framework for thermal–visible image registration, sensor fusion, and people tracking for video surveillance applications. *Computer Vision and Image Understanding* 116, 2 (2012), 210–221. https://doi.org/10.1016/j.cviu.2011.10.006

[43] M. Vandersteegen, W. Reusen, K. Van Beeck, and T. Goedemé. 2023. Person detection using an ultra low-resolution thermal imager on a low-cost MCU. In *Image and Vision Computing (IVCNZ 2022)*. Springer, 33–47. https://doi.org/10.1007/978-3-031-25825-1_3. MlDetection dataset, https://iiw.kuleuven.be/onderzoek/eavise/mldetection/home, by permission.

[44] Z. Tang, W. Ye, W.-C. Ma, and H. Zhao. 2023. What happened 3 seconds ago? Inferring the past with thermal imaging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR '23)*. IEEE. Thermal-IM dataset, BSD-3-Clause.

[45] M. Cokbas, P. Ishwar, and J. Konrad. 2020. Low-resolution overhead thermal tripwire for occupancy estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW '20)*. IEEE. https://doi.org/10.1109/CVPRW50498.2020.00052. TIDOS dataset, by permission.

[46] M. Lupión, A. Polo-Rodríguez, J. Medina-Quero, J. F. Sanjuan, and P. M. Ortigosa. 2024. 3D human pose estimation from multi-view thermal vision sensors. *Information Fusion* 104 (2024), 102154. https://doi.org/10.1016/j.inffus.2023.102154. Dataset: https://doi.org/10.17632/zsph6gvccz.1, by permission.

[47] K. Van Beeck, K. Van Engeland, J. Vennekens, and T. Goedemé. 2017. Abnormal behavior detection in LWIR surveillance of railway platforms. In *Proceedings of the 14th IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS '17)*. IEEE, 1–6. https://doi.org/10.1109/AVSS.2017.8078540. VIPER (Brugge) dataset, by permission, for scientific use only.

Appendix A: Choices made after seeing data

H1 to H3 followed Section 3's findings and several Section 6 analyses; Section 7.2 is the first test with rules fixed in advance, though its comparator, rule (b), was chosen on 13 October and its six-city verdict on 2 November, before the export. Section 3.2's measurements (1 to 8 September) overlap the three test days and informed the nowcast (weights fitted on 1 to 5 September). Section 6.1's 30-day window was scored after the three-day result was known. The display map, offset weight and original nowcast weights were chosen on earlier rows; the serving clamp and one-point push were fitted on the first 30% of the held-out cities' dates, inside the gate slice, and those cities gated every version from v2.2 on (Section 6.2). Of the "no event nearby" rows (56.1% of the corpus), 59.5% were defaults correlated with city, so Section 6.5 disables the events lookup for both arms. The persistence baselines, two-way bootstrap, age-mix standardisation, changed-value readings and test-window stickiness were computed on 13 October, after this paper's first version; rule (b), identified on the 30-day window and served from 13 October to 1 November, has no out-of-sample figure yet. The replication's curve-plus-latest-departure predictor had no pass rule. Owl's sealed tests: test 4's soccer sample grew from 98 to 393 crops after a frame-numbering fix, and test 5's bootstrap blocks were regrouped after first estimates (intervals only); every other plan change (tests 6 to 9, 11, 12 and 16, and LITIV 2012 after its count) preceded any prediction, and crowd-size comparisons were planned from test 7 on. For owl-5, the split was extended on 19 and 20 October and the rule between its candidates was amended three times on 2 November, all before any owl-5 training: BU-TIV joined both candidates, and the bootstrap units were coarsened from runs to recordings, then to each set's coarsest independent unit, then to one block for LTD; the gate's frame grid was fixed on 16 October, a day after its criteria.